



Bachelor Thesis

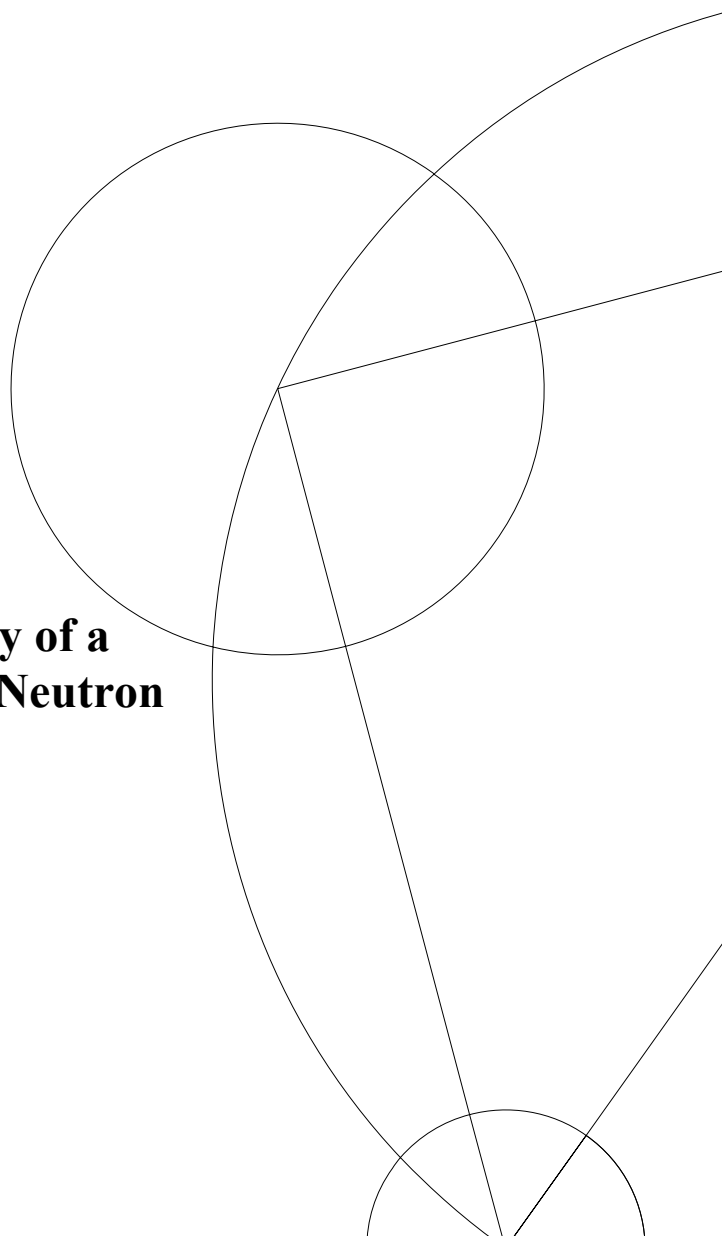
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Investigating the Optimal Geometry of a Vanadium Calibration Sample for Neutron Scattering

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Abstract

This work developed an independent minimalist neutron ray-tracing program McMini, which validated the Union framework in McStas, as of version 3.6.14, within 3×10^{-7} of the relative integrated intensity. McMini simulates multiple scattering within solid cylindrical and spherical samples. The comparison showed agreement in total intensity and angular scattering distributions. However, identical simulation instruments of a direct beam on a spherical and cylindrical monitor revealed monitor-related issues within McStas. The tests showed that a north-south asymmetry is present, where intensity is biased towards the northern hemisphere in the spherical monitor. This presents no limitation in the geometry optimization, since we focused on a small band on the equator and thus avoided the asymmetries at the edges. The cylindrical monitor recorded zero intensity in the highest angular bin column.

Several valid geometries were found for the hollow vanadium sample through multi-objective optimization. The multi-objective optimization located the Pareto solution set by minimizing a flatness metric and maximizing the total intensity within a 60° detector window. This optimization method found geometries in a range of radii, heights, and thicknesses, thus giving a range of usable geometries depending on experimental requirements. The optimized geometries showed less anisotropic spread and higher absolute intensity within the detector window.

Using the Born approximation to find an expected single-scattering intensity, we obtained a naive estimate of the scattered intensity from a hollow cylindrical vanadium sample. By comparing the observed scattered intensity to the expected single-scattering intensity, we found a self-absorption ratio. For the PSI vanadium sample with a height and radius of 10 mm and a wall thickness of 2.6 mm, this value was 30.2%.

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1 Introduction

Neutron scattering experiments can reveal the dynamics and structure of materials. Neutrons interact directly with atomic nuclei through the strong nuclear force and with magnetic moments through the electromagnetic force rather than the electron cloud. This allows neutron scattering to differentiate even light isotopes [1, p. 7]. Neutron simulation tools are useful for preparing for real experiments at spallation sources. Having a neutron scattering experiment simulated beforehand increases the efficiency of experimental design. The reliability of these tools is therefore crucial, and confidence that these tools correctly apply the physical properties will help design an improved calibration sample.

Vanadium is widely used in the calibration process because its scattering is dominated by incoherent processes. The key property is its relatively large incoherent scattering cross section and relatively small coherent scattering cross section, which leads to an approximately isotropic scattering distribution. However, even for vanadium with isotropic scattering, the measured angular distribution will be influenced by the geometry of the sample. Accurate calibration samples are needed to determine detector efficiencies, instrument resolutions, and systematic effects arising from the sample environment, thus motivating the need to understand the geometry-dependent scattering patterns.

McStas is a neutron ray-tracing software commonly used to model neutron scattering experiments. McStas allows for very detailed descriptions of neutron sources, optical components, detectors, and samples. Complex geometries and multiple scattering events within composite samples can be simulated within McStas using the Union framework [2]. Rather than modeling scattering components individually, multiple geometric objects can be grouped into a single physical sample with Union. This allows neutrons to scatter multiple times within a single object before escaping, using the Union framework and the `Incoherent_process` component [3]. Since multiple scattering can affect the measured intensity, it is important that the implementation is physically correct. Therefore, independent verification of the multiple scattering behavior is useful.

1.1 Project objectives

For this reason, we first developed a minimal Monte Carlo neutron transport program, McMini, written in the C programming language. This code was developed with the intention of creating a reference implementation to independently verify that McStas exhibits expected behavior when multiple scattering is simulated using the Union framework.

Secondly, agreement between the independent simulation code and McStas was evaluated using simple geometries such as hollow cylinders. Angular intensity distributions as well as scattering statistics were compared between the two implementations using the reduced chi-square χ^2_ν [4] and RMS residuals.

Lastly, once the simulation framework was validated, further simulations were performed in McStas. This program was then used to analyze the anisotropic scattering caused by hollow vanadium cylinders with different radii, heights, and wall thicknesses. The objective was to find geometries that reduce systematic errors to produce useful calibration behavior for neutron scattering experiments like those used at the CAMEA spectrometer at PSI in Switzerland.

2 Theoretical background

This chapter introduces and discusses the theoretical concepts required to understand the simulation methods and the physical processes that are important in neutron scattering. Fundamental concepts such as microscopic and macroscopic cross sections are reviewed along with the process of multiple scattering.

2.1 Scattering and absorption cross sections

The neutron scattering cross section determines how well a system scatters neutrons. It is expressed as [1, p. 12]

$$\sigma = \frac{1}{\Psi} \frac{dN}{dt}, \quad (2.1)$$

where σ has units of area and dN/dt has units of neutrons s^{-1} . The scattered intensity is normalized by the flux, so that σ is an intrinsic property of the system [1, p. 12]. At the single nucleus level, σ can be understood as the effective area that is perpendicular to the beam. It is also referred to as the microscopic scattering cross section [1, p. 12]. Since most experiments consider larger samples, the microscopic cross sections are converted to macroscopic cross sections. This can be accomplished by multiplying the number density of atoms in the material by the microscopic cross section [1, p. 13],

$$\Sigma_t = n\sigma_t, \quad (2.2)$$

where n is the number density of atoms. The macroscopic cross section has units of inverse length and represents the probability of interaction. Following the rules of probability, the total macroscopic cross section is additive, leading to

$$\Sigma_t = \Sigma_{\text{scat}} + \Sigma_{\text{abs}}, \quad (2.3)$$

where Σ_{scat} is the macroscopic scattering cross section, containing coherent and incoherent contributions, and Σ_{abs} is the macroscopic absorption cross section [1, p. 30]. Scattering lengths vary, but since we are dealing with macroscopic samples, the sample can average over all configurations (ensemble average). If we write the scattering length of nucleus j as

$$b_j = \langle b \rangle + \delta b_j, \quad (2.4)$$

then the coherent part of the scattering will be determined by the average scattering length $\langle b \rangle$, while the incoherent part is determined by the variance of the fluctuations. The total single nucleus scattering cross section is proportional to $\langle b^2 \rangle$, while the coherent contribution is proportional to $\langle b \rangle^2$. The difference is then the incoherent cross section,

$$\sigma_{\text{inc}} = 4\pi (\langle b^2 \rangle - \langle b \rangle^2) = 4\pi \langle (\delta b)^2 \rangle. \quad (2.5)$$

In the case of vanadium, the incoherent cross section is dominant, with the tabulated value [1, p. 16]

$$\sigma_{\text{inc}} = 5.08 \text{ barn}. \quad (2.6)$$

This is much larger than the coherent value $\sigma_{\text{coh}} = 4\pi b^2 = 0.0184$ barns [5]. Because the incoherent cross section dominates, vanadium has isotropic scattering with very weak Bragg peaks. The absorption cross section follows an inversely proportional relationship with velocity [1, p. 15], which will be important to model correctly in the simulation.

2.2 Neutron transport and Monte Carlo method

A key concept in Monte Carlo simulation is the sampling of free-flight distances using random numbers. The free-flight distances, s , represent the distance between interaction events. The probability that a neutron travels a distance s without interacting follows the Lambert-Beer law for attenuation [1, p. 31],

$$P(s) = e^{-\Sigma_t \cdot s}, \quad (2.7)$$

where Σ_t is the total macroscopic cross section of the material. The free-flight distances are therefore sampled as [6]

$$s = -\frac{\ln(\mathcal{U})}{\Sigma_t}, \quad (2.8)$$

where $\mathcal{U}$ is a uniformly distributed random number in $(0, 1]$. Random numbers are again used to decide whether an event is a scattering or absorption event. By applying this neutron transport logic for a large number of neutrons, statistical estimates of measurable quantities, such as intensity distributions, can be obtained.

2.3 Simulating neutron scattering

In neutron scattering experiments it is possible for a neutron to scatter multiple times before leaving the sample. These scattering events may deflect the neutron, transfer energy to the neutron, or absorb the neutron, altering the observed distribution. Depending on the geometry or cross section of the sample, multiple scattering can strongly affect the resulting signal [1, p. 7]. Calibration samples such as vanadium have a relatively large scattering cross section [1, p. 16] and can experience multiple scattering, altering the final scatter distribution and deviating from an ideal isotropic pattern [7]. In order to model neutron scattering, one must keep track of the neutron through sequential histories through the sample until it exits or is absorbed. This process lends itself to Monte Carlo simulations, since neutron histories can be tracked through individual interactions, where the location of the next collision can be sampled from the Lambert-Beer law for attenuation [8, 6]. Since parts of the code use biased or restricted sampling, it is useful to assign a weight to neutrons. The source may only target a finite area instead of sampling uniformly across all angles. The ray weight must account for the relevant phase-space factor so that ray intensity can be normalized correctly. Thus, p represents the Monte Carlo weight of that neutron history.

3 McMini methodology and validation of Union in McStas

3.1 Design philosophy and purpose

McMini is an independently developed minimal Monte Carlo neutron transport program implemented in C. McMini is simple by design, with the philosophy of reducing primary physical interactions to their core elements to best recreate the results of a real experiment. Its purpose is

to validate the results of the multiple scattering simulations performed with Union in McStas.

3.2 Source_simple model and implementation

The source implemented in McMini follows the same input parameters as the component Source_simple in McStas [3]. To compare the results of the two ray-tracing programs, they must produce statistically consistent neutron beams. For the source, that means consistent spatial, wavelength, and weight distributions. A neutron ray is described in the system by the trajectory

$$\vec{r}(s) = \vec{r}_0 + s\hat{u}, \quad (3.1)$$

where $\vec{r}_0$ marks the origin of the ray, $\hat{u}$ denotes the unit direction vector, and s is the ray parameter. Since our model ignores gravity and time-of-flight effects, we allow s to act as a measure of path length rather than physical time. Here, $\vec{r}_0$ is drawn from a uniform distribution on a circular source disk. The direction is sampled by choosing a point on a rectangular target area and normalizing the vector from the source point to the sampled target point. The wavelength is sampled uniformly according to

$$\lambda \sim \mathcal{U}[\lambda_0 - \Delta\lambda, \lambda_0 + \Delta\lambda]. \quad (3.2)$$

In the McMini implementation of Source_simple, the Monte Carlo weight, p of a neutron ray is factorized as

$$p = p_{\text{mul}} \cdot p_{\text{dir}}, \quad (3.3)$$

where

$$p_{\text{mul}} = \Psi 10^4 \pi R^2 2\Delta\lambda \quad \text{and} \quad p_{\text{dir}} = \frac{x_{\text{focus}} \cdot y_{\text{focus}}}{d^2} \cos^2 \theta. \quad (3.4)$$

In p_{mul} , Ψ is the source flux, the factor 10^4 is present to account for the unit conversion in the definition of the flux from cm^{-2} to m^{-2} , R is the radius of the circular source, and $\Delta\lambda$ is the constant used in the wavelength distribution. In p_{dir} , x_{focus} and y_{focus} are respectively the horizontal and vertical lengths of the rectangular target area, d is the distance between the sampled source point and the sampled target point in meters, and θ is the angle between the ray's direction and the normal to the source. As can be seen in equation 3.4, p_{dir} depends on the position sampled, since both d and θ will vary for each ray.

3.3 4π monitor implementation

The spherical 4π monitor with radius R records neutrons scattered in all directions and is therefore used as the primary monitor. The monitor uses the same angular coordinate convention as the 4π monitor in McStas, where $\theta \in [-90^\circ, 90^\circ]$ denotes latitude and $\phi \in [-180^\circ, 180^\circ]$ denotes longitude. The number of latitude bins is n_y and the number of longitude bins is n_x . The intersection point of a neutron ray with the sphere is found by geometrically describing the sphere with a quadratic equation and solving the equation for the s_{mon} -values,

$$(x_0 + \hat{u}_x s_{\text{mon}})^2 + (y_0 + \hat{u}_y s_{\text{mon}})^2 + (z_0 + \hat{u}_z s_{\text{mon}})^2 = R^2. \quad (3.5)$$

The angular coordinates θ and ϕ are found by taking the unit vector from the center to the hit position on the 4π monitor,

$$\theta = \text{atan2}\left(u_y, \sqrt{u_x^2 + u_z^2}\right), \quad \theta \in [-90^\circ, 90^\circ], \quad (3.6)$$

$$\phi = \text{atan2}(u_x, u_z), \quad \phi \in [-180^\circ, 180^\circ]. \quad (3.7)$$

The hit points are subsequently binned using the normalized bin indices,

$$f_\theta = \frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}}, \quad f_\phi = \frac{\phi - \phi_{\min}}{\phi_{\max} - \phi_{\min}}. \quad (3.8)$$

After that, the bin indices are computed and the neutron weight was added,

$$i_y = \lfloor f_\theta n_y \rfloor, \quad i_x = \lfloor f_\phi n_x \rfloor. \quad (3.9)$$

Finally, the weighted monitor signal per pixel that is reported is

$$I_{ij} = \frac{1}{N_{\text{count}}} \sum_{\text{hits}} p, \quad (3.10)$$

where p consists of the factors p_{mul} and p_{dir} as discussed in section 3.2, i and j denote the corresponding pixels, and N_{count} denotes the number of neutrons emitted from the source. No geometrical solid-angle normalization is applied internally. Thus, the raw monitor signal corresponds to the weighted sum of neutron probabilities per pixel.

Longitude vs. latitude binning. Using the definition of latitude and longitude bins on the 4π monitor, the data can be plotted on a rectangular map. This is convenient; however, equal steps in latitude and longitude are not equal in solid angle. In particular, the bins near the poles have a smaller solid angle than those near the equator. We therefore see that the rectangular map does not directly represent the intensity per solid-angle element. The differential solid angle when θ is the latitude angle is [9]

$$d\Omega = \cos \theta \, d\theta \, d\phi. \quad (3.11)$$

Thus, the solid angle between latitudes θ_1 and θ_2 is

$$\Delta\Omega = \int_0^{2\pi} \int_{\theta_1}^{\theta_2} \cos \theta \, d\theta \, d\phi = 2\pi (\sin \theta_2 - \sin \theta_1). \quad (3.12)$$

As a result, the bins located at the poles, ($\theta \rightarrow \pm 90^\circ$), represent smaller areas. A correction array of size $[n_x, n_y]$ is used to divide each bin by $\cos \theta$, where θ is the latitude angle in radians of that bin, approximately converting the intensity to solid-angle elements.

Comparison with McStas' 4π monitor. To validate the 4π monitor in McMini and McStas, the same settings were applied to a downstream simulation that contained only the source and the monitor. The simulation used 10^{11} neutron rays to ensure statistical precision in the comparison.

The direct beam footprint can be seen in figures 3.1a and 3.1b, where the same color scale

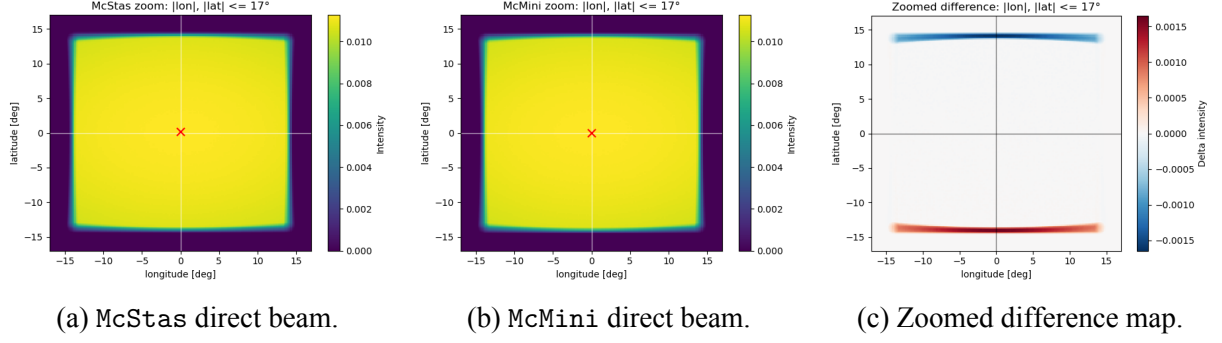


Figure 3.1: Comparison and zoomed difference map of the direct beam footprint on a 4π monitor between McStas and McMini. Red cross marks longitude and latitude centroid.

was used. Visually, the two direct beam distributions appear identical. However, if we subtract the McStas data from the McMini data, as done in figure 3.1c, there appears to be a systematic difference larger than statistical noise. The difference has local relative deviations of order $\sim 1.5 \times 10^{-2}$ compared to the intensity of the direct-beam. The difference lies close to the top and bottom edges of the direct beam footprint on the 4π monitor, not along the edge of the entire 4π monitor. The edge of concern is then the $\sim 15^\circ$ that the direct beam occupies on the 4π monitor. To test whether this difference was due to a north-south asymmetry, the southern hemisphere was mirrored into the northern hemisphere.

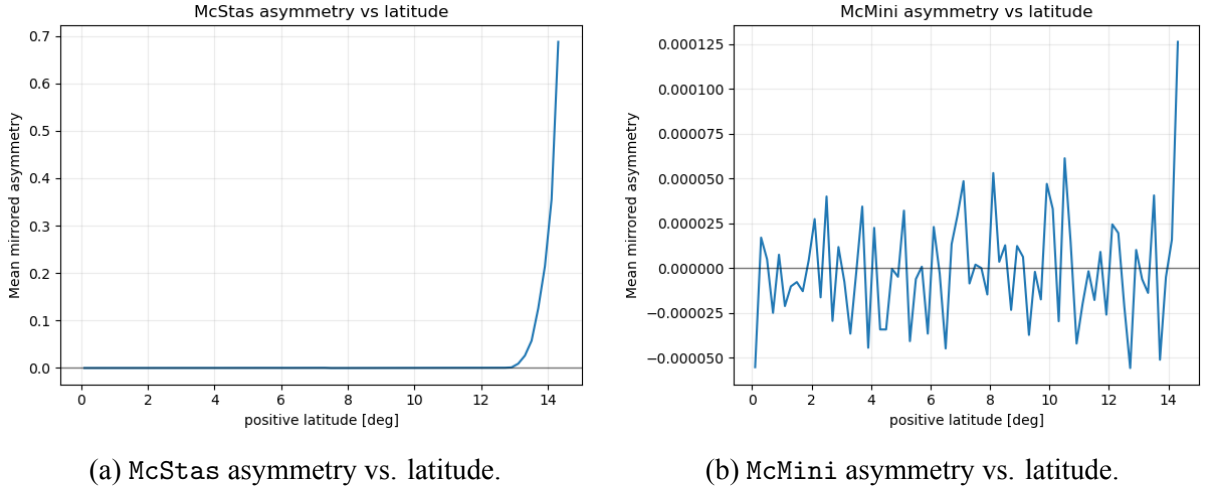


Figure 3.2: Mirrored direct beam asymmetry as a function of latitude for McStas and McMini.

Figure 3.2 shows that the McStas 4π monitor contains a systematic north-south asymmetry near the edge of the direct beam footprint. The asymmetry increases towards the angular boundary of the beam, at approximately 14° . In contrast, the corresponding McMini result shows no systematic north-south asymmetry, and the remaining difference is consistent with statistical noise. Interestingly, the issue was not present when testing the same setup on a flat plane monitor. Computing the degree of asymmetry with

$$A = \frac{|I_N - I_S|}{I_N + I_S}, \quad (3.13)$$

where I_N and I_S are the total intensities of the northern and southern hemispheres, McStas yields

7.30×10^{-3} and McMini yields 8.91×10^{-7} . This means that in McStas one hemisphere has 0.73% more intensity than the other compared to the total intensity, while McMini has essentially zero within statistical precision. The observed north-south asymmetry was investigated in detail. For the complete structural analysis of this asymmetry, see appendix C.

3.4 Cylindrical monitor bin offset

A cylindrical monitor was implemented and compared with the McStas component `Cyl_monitor_PSD` [3], where three issues were discovered. The rightmost angular bin column records zero intensity, there is a left-right asymmetry, and the top and bottom rows had an unexplained drop in intensity. The left-right asymmetry is likely caused by an angular bin shift and thus correlated with the missing rightmost bin column. A downstream beam onto the cylindrical monitor showed that the left-right asymmetry is

$$\frac{|I_L - I_R|}{I_L + I_R} = 3.60 \times 10^{-2}. \quad (3.14)$$

Shifting the monitor data one angular bin to the right ($\Delta b = +1$) and dropping the now-empty left edge bin restored symmetry:

$$\frac{|I_L - I_R|}{I_L + I_R} = 1.63 \times 10^{-6}. \quad (3.15)$$

Neither of the adjacent shifts correctly realigns the beam:

$$\Delta b = -1 : 5.12 \times 10^{-2}, \quad \Delta b = +2 : 3.60 \times 10^{-2}. \quad (3.16)$$

Therefore, $\Delta b = +1$ is uniquely the shift needed to minimize the asymmetry of the beam, approximately reducing its magnitude by 100%. Thus, it seems reasonable to conclude that the cylindrical monitor suffers from an angular displacement of one bin. The top and bottom rows had a large intensity reduction compared to the neighboring bins. The cause of this issue was not identified.

3.5 Sample implementation

Intersection logic. Computing the intersections with the cylinder is slightly more complicated than for the spherical monitor, since it has a finite height along the y -axis. The code finds intersection candidates and uses two conditions which are combined to compute the entry and exit points. Finding the intersections with the cylindrical wall and end caps requires solving these two equations, respectively,

$$(x_0 + \hat{u}_x s_{\text{cyl}})^2 + (z_0 + \hat{u}_z s_{\text{cyl}})^2 = R^2 \quad \text{and} \quad \pm \frac{H}{2} = y_0 + \hat{u}_y s_{\text{cyl}}, \quad (3.17)$$

where R is the radius of the cylinder and H is the height of the cylinder. Solving the two equations provides a total of four candidate values of s_{cyl} . Therefore, conditions are needed to select the two correct candidates of s_{cyl} , which represent the entry and exit points. A wall intersection is only valid if the y coordinate lies between the end caps, and an end-cap intersection is only

valid if the radial coordinate lies within the cylindrical walls. Written as the radial condition and the height condition,

$$x^2 + z^2 \leq R^2 \quad \text{and} \quad -\frac{H}{2} \leq y \leq \frac{H}{2}. \quad (3.18)$$

Using the two conditions in equation 3.18, the correct entry and exit points are selected and used to compute the path length $L = s_{\text{exit}} - s_{\text{entry}}$. From here the microscopic cross sections are converted to macroscopic cross sections, the absorption cross section is scaled by $\frac{2200}{v}$ [1, p. 15], and the free-flight distances are sampled.

Free-flight sampling. The path length L refers to how far the neutron ray travels inside the material. The free-flight distance, s , refers to how far the neutron ray travels before interacting, and it is sampled using equation 2.8. If the free-flight distance s is greater than the path length L , then the neutron ray is considered transmitted to the exit point of the surface. If the free-flight distance s is shorter than the path length L , then an interaction occurs within the material. The neutron ray is moved to the interaction point. The code then determines whether the event is scattering or absorption by comparing a random number with the probability that the event is a scattering event. In the general case, the total macroscopic cross section contains coherent scattering, incoherent scattering, and absorption as described in section 2.1. In the McMini implementation the coherent term is omitted,

$$P_{\text{scat}} = \frac{\Sigma_{\text{inc}}}{\Sigma_{\text{t}}^{\text{McMini}}}, \quad (3.19)$$

where Σ_{inc} is the macroscopic incoherent scattering cross section and $\Sigma_{\text{t}}^{\text{McMini}} = \Sigma_{\text{inc}} + \Sigma_{\text{abs}}$. Using the tabulated microscopic cross sections and the number density for vanadium gives $\Sigma_{\text{inc}} = 0.367 \text{ cm}^{-1}$ and $\Sigma_{\text{t}}^{\text{McMini}} = 0.734 \text{ cm}^{-1}$ at the reference thermal velocity $v = 2200 \text{ m s}^{-1}$ [1, p. 16]. If the random number is larger than P_{scat} , the neutron ray is killed and discarded. Otherwise, it is classified as a scattering event. The code then samples a new direction by sampling uniformly from the unit sphere.

Multiple scattering algorithm. The code handles multiple scattering using the function `scatterer_cyl_interact`. It computes the intersections with the cylinder, samples free-flight distances, and decides whether the event is scattering or absorption, updates the neutron ray's position, and returns the event type. As long as the neutron ray is alive, this function is called repeatedly. In McMini, if the neutron ray is absorbed, it is removed completely. This differs from McStas, which uses weighted absorption, where a neutron ray stays in the system but its weight is reduced by its chance of being absorbed.

3.6 Solid vanadium cylinder validation

Validation setup and metrics. To validate the Union framework in McStas, we compared the simulated scattering patterns with those obtained from McMini. The comparison includes intensity cuts along the equator and meridian to investigate the angular dependence of the scattering signal. We investigate the integrated intensities and also the bin-by-bin intensity difference

between McStas and McMini, defined as

$$\Delta I = I_{\text{McMini}} - I_{\text{McStas}}. \quad (3.20)$$

These metrics were quantified using statistical metrics such as residual RMS, reduced chi-square (χ^2_ν), and the Pearson correlation.

Equator and meridian cuts. The equator cut is taken as the integrated intensity in the 4° latitude band centered on the equator, $\theta \in [-2^\circ, 2^\circ]$, plotted against longitude ϕ . Similarly, the meridian cut is taken as the integrated intensity in the 4° longitude band centered at the meridian, $\phi \in [-2^\circ, 2^\circ]$, plotted against latitude θ . The raw output of the 4π monitor is binned using latitude-longitude bins. The cosine correction described in section 3.3 was applied separately to each latitude row. For this reason, both the raw intensity I and the cosine-corrected intensity $I/\cos\theta$ are reported separately for the 4π monitor validation. Figure 3.3a shows the equator band cut and figure 3.3b shows the meridian band cut.

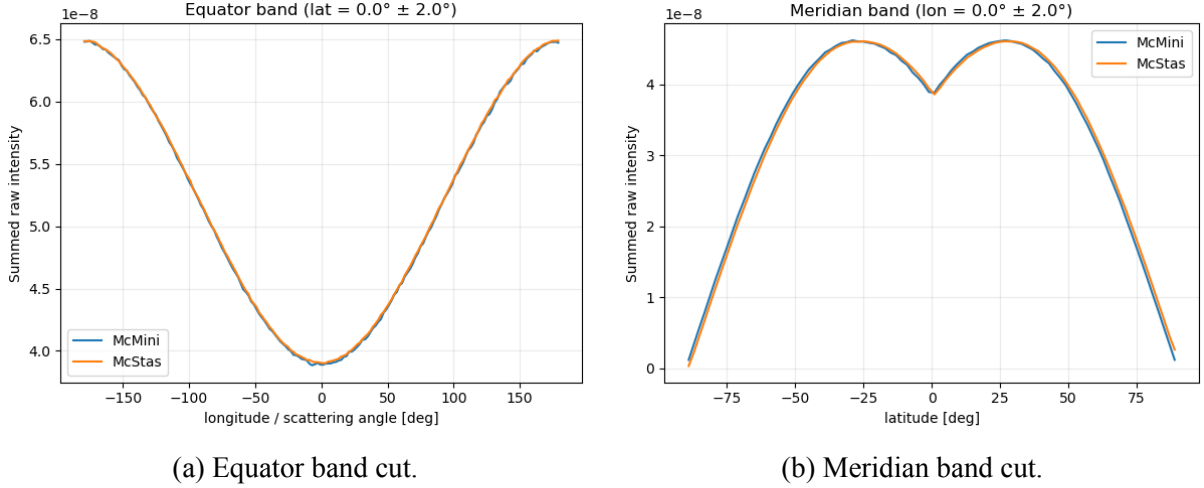


Figure 3.3: Equator and meridian band cuts for the solid cylinder case.

Overall, the equator band cut shows good agreement between McStas and McMini. As shown in table 3.1, the raw equator cut Pearson correlation is 0.999961, and the ratio of integrated sums is 0.997910. The reduced chi-square of the raw equator cut is $\chi^2_\nu = 3.02$, which is larger than unity. However, at these high neutron counts, even small systematic residuals that are approximately flat over many bins appear significant in the χ^2 pull distribution. Therefore, we compute the normalized shape comparison, resulting in a shape-only reduced chi-square of 1.74×10^{-8} . This demonstrates that the angular dependence of the equator cut is reproduced very accurately.

Integrated intensities and residuals. The validation results are shown in table 3.1. The integrated intensity ratio for the uncorrected map is $1 + 3 \times 10^{-7}$, which shows that the total intensity agrees to high precision. This is solid evidence that the scattering probability and neutron weights are implemented consistently.

Quantity	Result
Raw integrated intensity ratio	$I_{\text{McMini}}/I_{\text{McStas}} = 1 + 3 \times 10^{-7}$
Residual RMS, raw map	3.19×10^{-10}
χ^2_ν of full difference map	245
McMini raw north-south asymmetry	3.07×10^{-6}
McStas raw north-south asymmetry	1.47×10^{-2}
Raw equator-cut Pearson correlation	0.999961
Raw equator-cut sum ratio	0.997910
Raw equator-cut χ^2_ν	3.02
Raw equator-cut shape-only χ^2_ν	1.74×10^{-8}

Table 3.1: Validation summary comparing McMini and McStas for the solid vanadium cylinder.

Agreement levels. This validation shows excellent agreement in the integrated intensity ratio between the two neutron transport programs. The bin-level residuals are small in absolute value, with a residual RMS of 3.19×10^{-10} showing a typical per-bin mismatch of $\sim 10^{-10}$. The Pearson correlation and sum ratio close to unity show strong agreement in the angular dependence of the scattered intensity.

Remaining discrepancies. The $\chi^2_\nu = 245$ from the full difference map is consistent with the north-south asymmetry. This is the most significant systematic disagreement between McStas and McMini in the McStas 4π monitor. McStas presents 1.47% more intensity in one hemisphere compared to the total intensity, while McMini has $3.07 \times 10^{-4}\%$. That is, McMini remains nominally symmetric within uncertainties, while McStas shows a systematic redistribution of intensity between the north and south hemispheres. The asymmetry was observed in the simulation of the solid vanadium cylinder and in the extra validation of the solid vanadium sphere, see appendix B.2. This indicates that the effect is not geometry dependent but an artifact that increases as $\theta \rightarrow \pm 90^\circ$. The asymmetry was visible in the reduced χ^2 , as it was far above unity. However, all cases showed the same total integrated intensity within statistical uncertainty, indicating a redistribution error. Diagnostics revealed that the problem originated in part from the PSD_monitor_4PI component. Tests with a direct beam on the 4π monitor still showed asymmetry, while a direct beam on the flat plane monitor showed no asymmetry.

Confidence in Union implementations. The simplicity of McMini is crucial for confidence in the validity of the simulation results. McMini implements only the necessary physics, such as microscopic scattering and absorption length, unit cell volume, and wavelength-dependent absorption. The agreement of ray propagation through simple geometries, apart from the observed north-south asymmetry, supports the implementation. McMini does not account for absorption by weight adjustment as in McStas and therefore presents a higher level of noise. This strengthens the validation because it means that when treating the absorption of neutron rays in a fundamentally different way, we still obtain the same result. This is strong evidence that the Union framework correctly implements cross sections. The only difference observed between McMini and McStas that cannot be attributed to noise is the north-south asymmetry. The Union framework has been validated and is consistent with the physical models of cross sections. However, the north-south asymmetry that we have observed, and the missing highest angular bin column in the cylindrical monitor, should be kept in mind when interpreting monitor data.

4 Scattering quality and geometry optimization in McStas

Simulation setup in McStas. We were given a full instrument file from PSI that replicates the instrument monitors, geometry, and sample parameters of the CAMEA instrument [10]. The built-in detectors, which were helium-filled tubes and analyzer crystals, required a lot of computational power. They were replaced with a spherical monitor with the vanadium sample at the center. The monitor was rotated such that the vanadium cylinder's axis of rotation aligned with the poles. The simulation used the default parameter values given in the instrument file. The default values can be seen in table D.1 in the appendix. The 4π monitor had a radius of 1 m and 180×90 angular bins. The monitor used the `WHEN(raw_scattered > 0)` condition. This means that the monitor did not measure the direct beam, but only neutrons that had scattered at least once.

4.1 Scattering quality metrics

Key metrics were assigned to a simulation to quantify the anisotropic spread of the sample. These metrics were used in combination with each other and separately.

Summed intensity. The first metric is the total intensity of all bins within two symmetric windows centered on the equator ($\theta = 0^\circ$) centered left and right in the scattering plane at longitude $\phi = \pm\phi_0$. The window has a longitude width of 60° and a latitude width of 4° , chosen to match the detectors used at PSI. We define the sum as follows:

$$I_{\text{tot}} = \sum_{(\phi_j, \theta_i) \in W} I(\phi_j, \theta_i), \quad (4.1)$$

where W denotes the set of all bins within the two symmetric windows. This metric is useful when comparing intensities generated by geometries different from the original CAMEA sample.

Reduced chi-square. The second metric is the reduced chi-square (χ_ν^2) [4] obtained by fitting a constant, c , to the latitude-averaged bin intensity value inside the selected windows.

$$\chi_\nu^2 = \frac{1}{N_\phi - 1} \sum_{j=1}^{N_\phi} \frac{(\bar{I}_j^{\text{band}} - c)^2}{\sigma_{\bar{I},j}^2}. \quad (4.2)$$

One degree of freedom is lost from fitting with one parameter. The mean intensity and variance are defined as

$$\bar{I}_j^{\text{band}} = \frac{1}{N_\theta} \sum_{|\theta_i| \leq \theta_0} I(\phi_j, \theta_i), \quad \sigma_{\bar{I},j}^2 = \frac{1}{N_\theta^2} \sum_{|\theta_i| \leq \theta_0} \sigma^2(\phi_j, \theta_i). \quad (4.3)$$

The χ_ν^2 metric shows how well a constant, c , can be fitted, allowing us to quantify the level of anisotropy within the two symmetric windows.

Coefficient of variation. The coefficient of variation is used to understand the relative variability of the mean intensity [11]. It is simply defined as

$$CV = \frac{\sigma_I}{\bar{I}}. \quad (4.4)$$

The coefficient of variation is a relative measure that allows us to quantify the standard deviation in relation to the mean [11]. It works well in conjunction with χ^2_ν because it is less affected by higher bin uncertainties due to lower statistics.

Expected single-scattering intensity. We define the expected single-scattering intensity, hereafter referred to as the expected intensity, $I_{\text{exp}}^{(1)}$, from the differential scattering cross section [1, p. 13]. This provides a naive first-order estimate of the scattered intensity by assuming incoherent isotropic scattering and ignoring effects such as absorption, attenuation, and multiple scattering.

$$I_{\text{exp}}^{(1)} = \Psi N \sigma_{\text{inc,V}} \frac{\Delta\Omega}{4\pi} \quad (4.5)$$

where N is the number of atoms illuminated by the beam and it is found using $N = nV$, where n is the number density of vanadium and V is the illuminated volume of the sample. The incoming flux density Ψ is estimated from a monitor placed at the sample position and integrating the incoming intensity over the projected sample area and dividing by that same area. $\sigma_{\text{inc,V}}$ is the incoherent microscopic scattering cross section of vanadium. The solid angle $\Delta\Omega$ is computed for a bin $2^\circ \times 2^\circ$ using equation 3.12 and multiplied by the number of bins in the two symmetric windows to find the total angular region. In a thin sample where self-absorption and multiple scattering can be ignored, the ratio

$$R = \frac{I_{\text{obs}}}{I_{\text{exp}}^{(1)}}, \quad (4.6)$$

should approach unity. The deviations from unity quantify the effect of neutron rays being self-absorbed or scattered multiple times.

4.2 Observed to expected intensity in the thin wall thickness limit

For thin walls, a small fraction of neutron rays should be absorbed and in the limit the ratio should approach unity. We used the PSI instrument file and default instrument parameters, except for decreasing wall thicknesses down to 10^{-3} mm. The uncertainties in the ratio were computed from the uncertainty in the observed and expected values. The uncertainty in the observed intensity was determined by summing the variances of each bin and taking the square root. The uncertainty in the expected intensity comes from the uncertainty in the incoming beam,

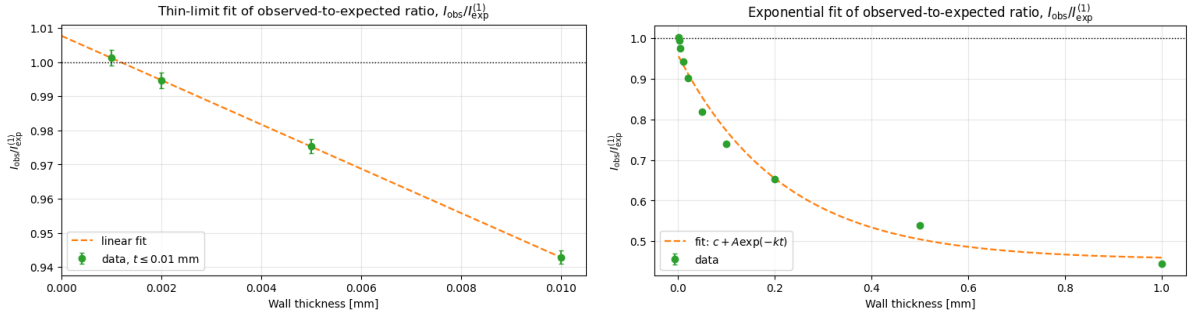
$$\sigma_O = \sqrt{\sum_i \sigma_{O_i}^2} \quad \text{and} \quad \sigma_E = E \frac{\sigma_\Psi}{\Psi}, \quad (4.7)$$

where σ_O is the uncertainty in the observed intensity within the two symmetric windows and σ_E is the uncertainty in the expected intensity within the two windows. Using standard error

propagation, we get the final uncertainty on the ratio of

$$\sigma_R = R \sqrt{\left(\frac{\sigma_O}{O}\right)^2 + \left(\frac{\sigma_E}{E}\right)^2}. \quad (4.8)$$

Note that these uncertainties only include the uncertainties of the statistical monitor. Uncertainties from effective illuminated volume, sample size, cross-section values, and choice of flux integration area have not been included.



(a) Observed intensity over expected intensity in the thin-limit of wall thickness.

(b) Observed intensity over expected intensity as a function of wall thickness.

Figure 4.1: Comparison of the thin-limit intensity ratio

The linear fit to the points with (thickness ≤ 0.01 mm) shown in figure 4.1a gives the parameter values

$$R(t) = 1.0077 - 6.4787 \cdot t, \quad (4.9)$$

where t is the thickness of the wall in mm. For larger wall thicknesses, the ratio decreases below unity because the expected intensity scales with the amount of illuminated vanadium, while the simulation includes attenuation, absorption, and multiple scattering. These effects reduce or redistribute the intensity reaching the selected angular windows. The full thickness dependence was fitted with an exponential model.

$$R(t) = c + A \exp(-kt), \quad (4.10)$$

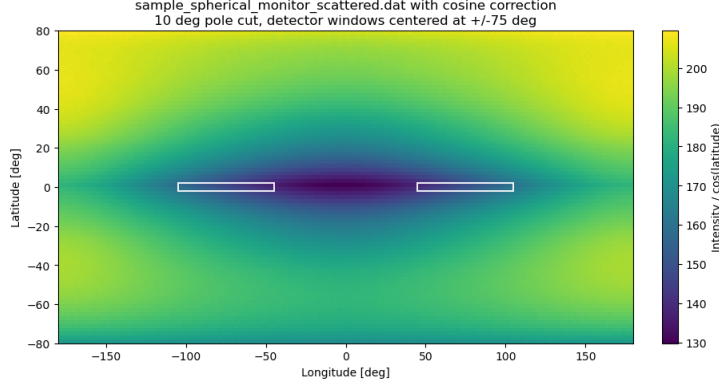
The fitted parameters were:

$$c = 0.4545, \quad A = 0.5032, \quad \text{and} \quad k = 4.6175 \text{ mm}^{-1}, \quad (4.11)$$

giving an extrapolated value at $t = 0$ mm of $R(0) = c + A = 0.9577$. The exponential fit describes the overall decrease with wall thickness, but the linear thin-limit fit is more appropriate when estimating the thin-limit value, since it only uses the points closest to $t = 0$ mm. The reason why the exponential model fitted poorly was not located.

4.3 Full instrument test with the CAMEA vanadium sample geometry

A high statistics simulation was performed with the CAMEA hollow cylinder sample geometry, shown in table 4.1, using the instrument file provided by PSI [10] with 10^{11} neutron rays. The spherical monitor was corrected for latitude-dependent solid angle. The regions within 10° of the poles were removed to avoid numerical singularity and to reduce the north-south asymmetry. The heatmap is shown in figure 4.2.

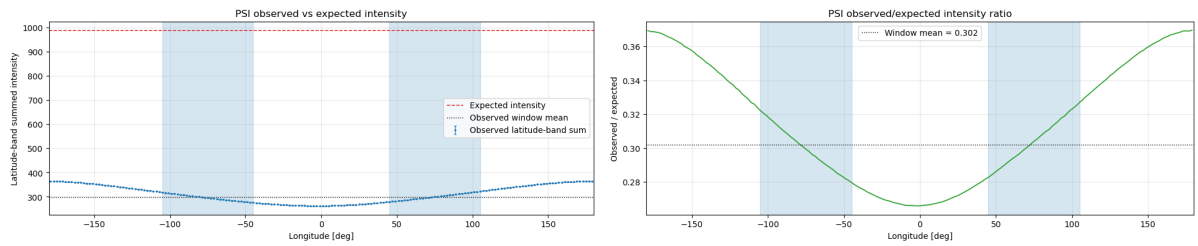


Parameter	Value
Radius, R	10 mm
Height, H	10 mm
Wall thickness, t	2.6 mm

Table 4.1: CAMEA vanadium sample geometry used in the full instrument test.

Figure 4.2: Heatmap obtained when simulating the CAMEA vanadium sample in the PSI instrument. The white boxes indicate the equatorial latitude bands used later.

The heatmap shows a large intensity reduction in the forward beam direction. The asymmetry, computed using equation 3.13, is $A = 1.49\%$, which is very close to the degree of asymmetry computed in section 3.6. We extract the latitude-summed intensity along the equator to better understand what happens at the equator line. Comparing the latitude-summed intensity to the expected intensity, we find the ratio (equation 4.6) but plotted as a function of longitude.



(a) Equatorial latitude integrated intensity compared to the expected intensity across longitude. (b) Angular ratio from equation 4.6, plotted as a function of longitude.

Figure 4.3: Comparison of the angular intensity distribution with the expected intensity and the corresponding observed-to-expected ratio.

The observed intensity compared with the expected intensity is shown in figure 4.3a and the ratio as a function of longitude is shown in figure 4.3b. The analytical summed intensity within the two $60^\circ \times 4^\circ$ windows and the observed intensity were, respectively,

$$I_{\text{exp}} = 61285.3 \quad \text{and} \quad I_{\text{obs}} = 18509.0 \pm 0.9. \quad (4.12)$$

This gives an observed to expected intensity ratio of $\sim 30.2\%$. The reduction occurs because the expected intensity only accounts for single scattering events and assumes that all neutrons are scattered isotropically. The ratio can be interpreted as an effective self-absorption factor for the selected angular window. We now evaluate the scattering produced using the quality metrics discussed in section 4.1. The mean intensity within an equatorial latitude band and its corresponding standard deviation were

$$\bar{I} = 298.532 \pm 0.014 \quad \text{and} \quad \sigma_I = 13.12, \quad (4.13)$$

giving a coefficient of variation of:

$$CV = 0.0439577. \quad (4.14)$$

The reduced chi-square value obtained by fitting a constant c to the two windows was:

$$\chi^2_\nu = 13317. \quad (4.15)$$

Together, the coefficient of variation and the reduced chi-square show that assuming isotropic spread within the detector arm is not valid. We can further quantify this deviation by looking at the peak-to-peak variation and the maximum absolute deviation from the mean. The peak-to-peak variation relative to the mean and the maximum absolute deviation were, respectively,

$$\frac{I_{\max} - I_{\min}}{\bar{I}} = 0.156791 \quad \text{and} \quad \frac{\max |I - \bar{I}|}{\bar{I}} = 0.082544, \quad (4.16)$$

demonstrating that the residual structure across the window is significantly larger than expected from counting uncertainty alone. The windows show clear anisotropy with a high reduced chi-square from a mean fit, and a peak-to-peak difference of $\sim 15.7\%$ and a maximum absolute deviation of $\sim 8.25\%$.

4.4 Optimization study and recommended geometry for calibration use

The objective of this chapter is to identify vanadium sample geometries that provide a strong calibration signal while remaining as close to an isotropic scattering distribution as possible. This requires studying not only the scattered intensity but also the angular flatness of the scattering pattern and the deviation from the expected intensity. The optimized geometry is identified using the scattering quality metrics discussed in section 4.1 applied to simulations performed using Union in McStas. The beam at the CAMEA instrument places a constraint on the sample height of 10 mm. Initially, this constraint is ignored to locate a generally optimized geometry, and after that the constraint will be considered.

Pareto optimization. Multi-objective optimization, or Pareto optimization, is a branch of optimization that suits our goal. It is used when more than one objective is to be minimized or maximized simultaneously. In this case, the flatness metric and the total intensity. The answer is a set of possible solutions that give the best trade-off between competing objectives [12, p. 2]. In a single objective search, the answer is easily determined by comparing the objectives to each

other. For multi-objective searches, the goodness of a solution is instead determined by dominance. Dominance is defined such that the solution x_1 dominates x_2 if x_1 is not worse than x_2 in all objectives or x_1 is strictly better than x_2 in at least one objective [12, p. 5]. The non-dominated solution set is the set of all solutions that are not dominated by any other solution. This non-dominated solution set is called the Pareto optimal solution set. The boundary around the Pareto solution set forms the Pareto front [12, p. 7].

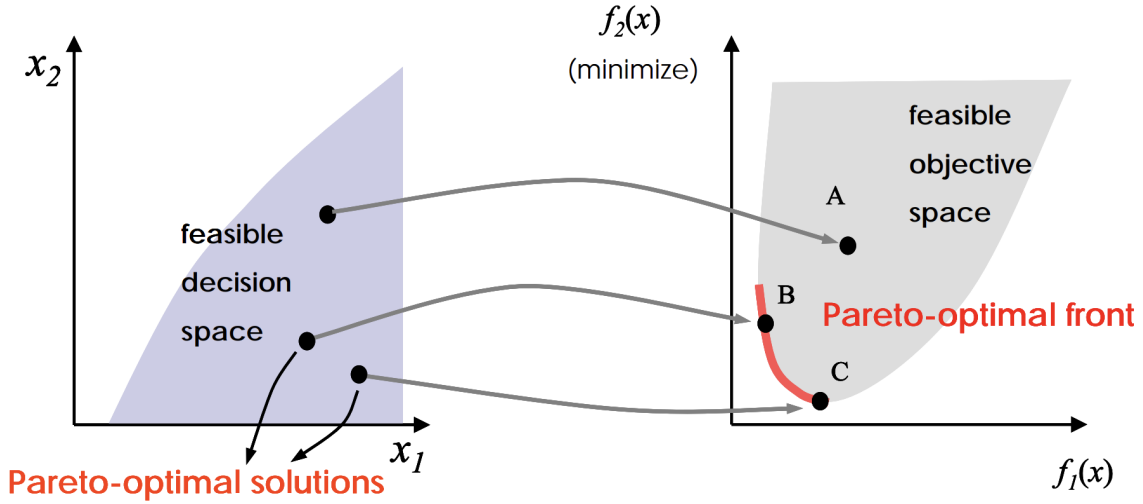
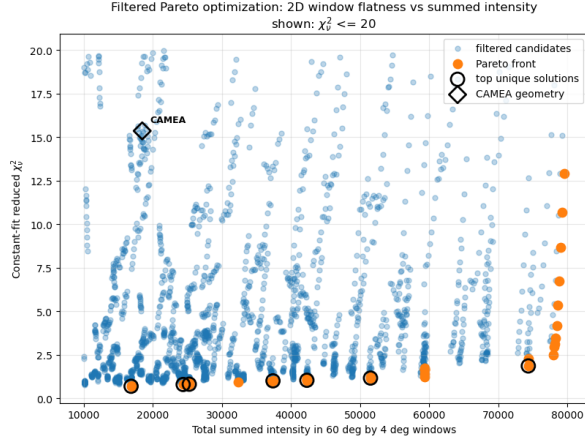


Figure 4.4: Illustration of the mapping from feasible solutions in the decision space to the objective space. Reproduced from [12, p. 8].

Pareto optimization with no sample constraints. By ignoring the sample constraint, we can investigate samples in a wider range of geometries that might be more favorable. Simulations were performed in a wide range of heights, radii, and thicknesses and should provide a good estimate of the best geometry. The geometry ranges were $H = [5, 50]$ mm, $R = [5, 15]$ mm, and $T = [0.5, 3.0]$ mm for a total of 270 different geometries. The two symmetric windows were allowed to move up to 20° left and right in increments of 2° , which gave a set of 2981 possible solution candidates. It makes sense to look for unique solution sets, since the Pareto solution set can give solutions that are alike. By unique, we mean that if a sample with the same radius and height but multiple different thicknesses or window offsets is found in the Pareto optimal solution set, then we only choose one of them. Simulations that give a total intensity within the two symmetric windows of $I < 10^4$ are filtered out. The cutoff was chosen because the CAMEA sample has $I_{\text{obs}} = 1.85 \times 10^4$ within the windows and we do not want to consider geometries with much lower intensity than that. The simulations that yielded $\chi_\nu^2 > 20$ were also removed.

After computing χ_ν^2 and the total intensity within the two windows, multi-objective optimization was performed to find the non-dominated solution set, shown in figure 4.5. Solutions with intensity $I < 10^4$ and $\chi_\nu^2 > 20$ were filtered out. The seven non-dominated solutions sets that are unique in height and radius are shown in table 4.2. We observe that some of the geometries give $\chi_\nu^2 < 1$. In general, this can suggest that the uncertainties are too large. However, values of χ_ν^2 less than one are not surprising in this study because the uncertainties are estimated from finite Monte Carlo simulations. Only 2×10^8 neutron rays were used to estimate the inten-



Rank	r [mm]	h [mm]	t [mm]	Offset [deg]	χ^2_ν	Sum [10^4]
1	5.00	35.00	1.00	-8.0	0.701	1.69
2	7.50	30.00	1.00	-12.0	0.803	2.44
3	7.50	40.00	1.00	-4.0	0.819	2.53
4	7.50	50.00	2.00	+8.0	1.01	3.74
5	10.00	40.00	1.50	-8.0	1.04	4.23
6	12.50	50.00	1.50	-12.0	1.17	5.15
7	15.00	50.00	2.50	-20.0	1.87	7.43
CAMEA	10.00	10.00	2.60	+0.0	15.4	1.85

Table 4.2: Top unique Pareto solutions after filtering, with the current CAMEA geometry shown for comparison.

Figure 4.5: Solution set for various geometries.

sity for each geometry. The resulting statistical estimates of intensity and uncertainty are only approximate. Therefore χ^2_ν should be used more as a relative measure of flatness than as an absolute goodness-of-fit criterion. The 36 non-dominated solution sets shown in figure 4.5 as orange points form the Pareto front, representing the best compromise between angular flatness and total intensity. The seven unique solution candidates are marked by a black circle and they are ranked on their χ^2_ν . The result obtained from the CAMEA vanadium sample is marked by a square, and its placement indicates that it is not optimal, both in flatness and in total intensity. The Pareto solution set located on the lower part of the Pareto front displayed very little anisotropy within the two symmetric windows. The expected intensity for the optimized sample geometries could not be computed. This is because the beam did not cover the entire sample, and the non-uniformity of the beam made it too difficult to estimate the flux density Ψ consistently.

Pareto optimization with sample constraints. The geometry at CAMEA requires the sample to be at most 10 mm tall to ensure the proper detector efficiency. Simulations were performed with fixed height and radius, both of 10 mm, across wall thicknesses ranging from 1.0 mm to 3.0 mm. Fewer geometries were tested, so the number of neutron rays was increased to 2×10^9 for better statistics. For this scan, the detector center was fixed at $\phi \pm 75^\circ$.

The Pareto optimization yielded the plot in figure 4.6 and the geometry for the sample with the highest summed intensity is shown in table 4.3. The plot shows that the intensity within the windows increases with thickness, but only up to a point. For wall thicknesses greater than 3 mm, the intensity starts to decrease, because scattered neutrons must travel through more material. Furthermore, as the wall thickness increases, the anisotropy increases as well. Therefore, a thinner wall would be beneficial, as the maximum deviation in the window decreases from 0.0825 to 0.0578, while having an intensity, close to what CAMEA yielded, of 1.46×10^4 in the windows. The plot indicates that there is not one specific ideal thickness, but rather a range of usable thicknesses. The Pareto optimization identified seven thicknesses near the bottom of the curve, ranging from 0.7 mm to 1.25 mm.

The intensity distribution and the observed-to-expected intensity as a function of longitude for the sample with $t = 1.25$ mm are shown in figure 4.7. The mean ratio within the windows is 41.3%. The benefit is small, so if possible, a taller sample would be much preferable, since a taller sample displays less anisotropy and has close to or higher absolute intensity.

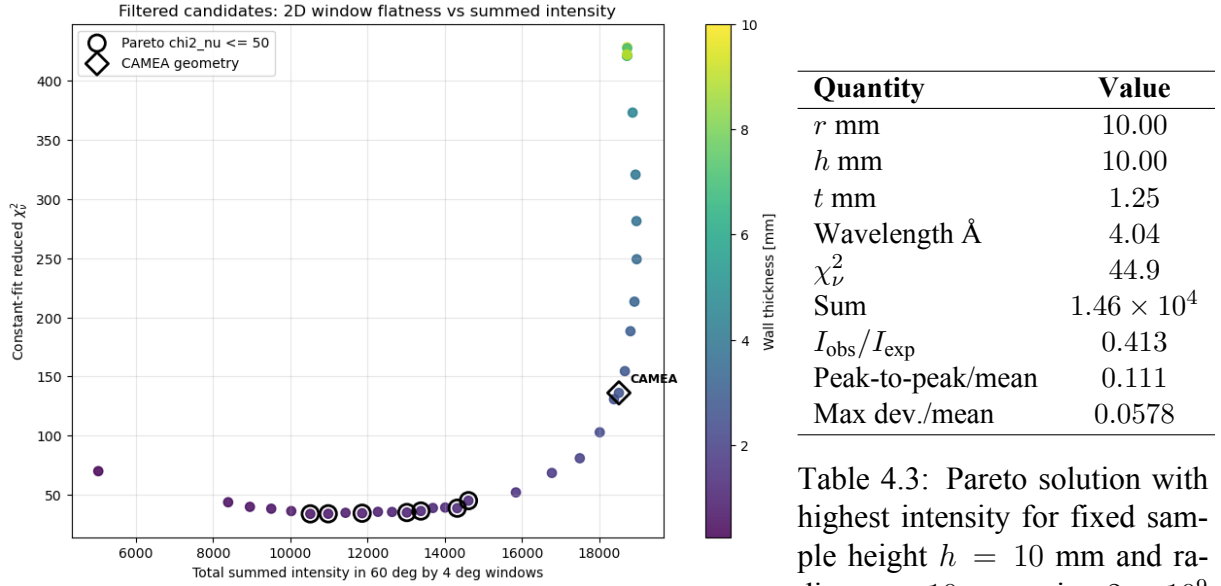


Figure 4.6: Pareto optimization with a sample of fixed height simulated neutrons and radius.

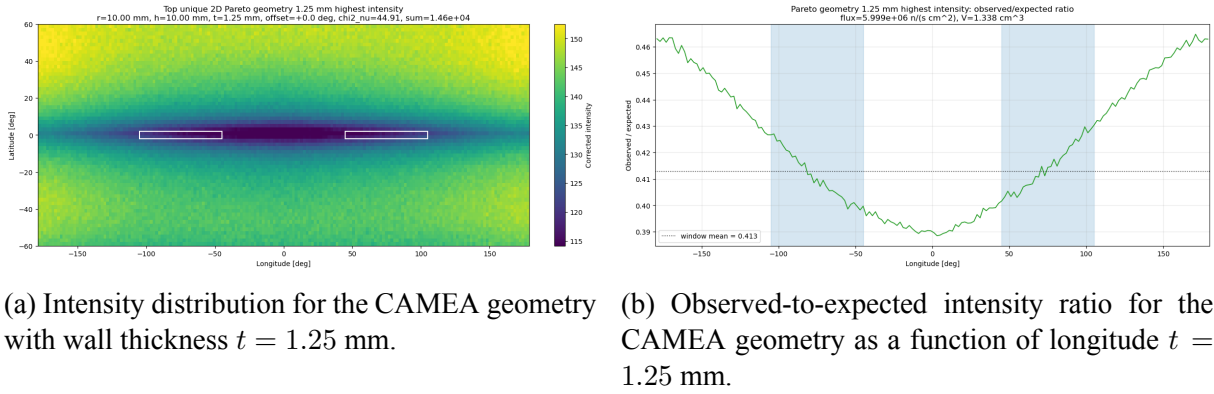


Figure 4.7: Intensity distribution and observed-to-expected intensity as a function of longitude.

5 Discussion and conclusion

Limitations of Monte Carlo model. First, we discuss the results and observations made in chapter 3. While McMini runs as a pure Monte Carlo physics simulation, McStas weighs neutron ray absorption, rather than removing absorbed rays, as a variance reduction technique. This contrast is the biggest difference between the two engines. McMini works as a minimalist validation tool that removes the absorbed neutron rays. This results in larger statistical noise, which will affect the reliability of the equator cut and meridian cut. However, since McMini and McStas treat neutron rays in a fundamentally different way, this result supports the validity of the Union framework. It was demonstrated that treating neutron rays in the manner of a weight absorption scheme produces statistically consistent results with a complete absorption of neutron rays. Agreement was not proven for every possible configuration of Union. Agreement was shown for the geometries and observables tested. This is solid evidence that the implementation is correct for the scattering orders, geometries, and monitor quantities checked.

Statistical precision. The integrated intensity ratio of McMini to McStas was $I_{\text{McMini}}/I_{\text{McStas}} = 1 + 3 \times 10^{-7}$ as seen in table 3.1. Thus, the two ray-tracing programs agree within 3×10^{-7} in their relative integrated intensity. This level of agreement is consistent with statistical noise up to the uncertainty of the simulation. The large χ^2_ν is consistent with the observed north-south asymmetry and shows that there is detectable bias in the Union implementation. However, considering the Pearson correlation close to unity in the equator cut, table 3.1, the large χ^2_ν becomes more subtle, and indicates that the physical process is implemented consistently.

Monitor-related limitations. The main limitation observed in McStas was the north-south asymmetry in the PSD_monitor_4PI component. The asymmetry levels for the cylindrical vanadium sample were 1.47×10^{-2} , see section 3.6, and for the spherical vanadium sample they were 1.43×10^{-2} , see appendix B.2. This shows that the asymmetry was consistent between geometries. Furthermore, analysis of the asymmetry with only a direct beam onto a flat monitor and a spherical monitor showed that the asymmetry was present only for the spherical monitor. This leads us to believe that it is a problem specific to the 4π monitor. The consequence of this likely monitor-related artifact is that it compromises detector efficiency near the poles. When using simulated data recorded with the PSD_monitor_4PI component, data near the poles should be treated with caution. The Cyl_monitor_PSD component has a binning issue as discussed in section 3.4. The column of the rightmost bin records zero intensity, which causes the intensity to be asymmetric from left to right.

Thin limit. We now discuss the results of chapter 4 and what we inferred from the CAMEA simulation and Pareto optimization. Simulating the CAMEA sample with decreasing wall thicknesses showed that the observed intensity approached the expected intensity, after which the ratio decreased rapidly with increasing thickness, reaching 44.5% at a wall thickness of 1 mm. Using the CAMEA instrument, we simulated the sample with decreasing wall thicknesses to investigate the thin wall limit. The result is shown in figure 4.1b. For samples with large wall thicknesses, we expect a larger deviation from the expected intensity because neutron rays have to travel through more material and therefore have a greater chance of being absorbed. When approaching very thin samples, we would thus expect very little deviation because all scattered neutrons have a very small escape path, and the ratio $R = I_{\text{obs}}/I_{\text{exp}}^{(1)}$ should approach unity. The thinnest sample of 10^{-3} mm yielded $R = 1.0013$, the extrapolation with a linear fit yielded $R(0) = 1.0077$, and the extrapolation with the exponential fit yielded $R(0) = 0.9577$. The only difficulty in the expected intensity was estimating the flux density at the sample because of the non-uniform spread of the beam.

Full CAMEA instrument test. The high statistics simulation with the exact CAMEA instrument sample geometry showed a large angular dependence and a large intensity decrease in the equatorial band. The test demonstrated that the CAMEA sample is not ideal if the goal is to reach low anisotropic spread. Although vanadium is characterized by a dominant incoherent scattering cross section, the resulting anisotropy is, in practice, a product of phenomena such as self-absorption and multiple scattering, which introduce angular dependence in the observed angular distribution. It is therefore necessary to correctly model the anisotropy produced by samples of various geometries. The vanadium sample from the CAMEA instrument showed

the least intensity in the beam direction as seen in figure 4.2. The ratio of the observed to the expected intensity within the two symmetric windows was $R = 0.302$. The ratio varies with longitude: Along the beam direction, we observed $\sim 27\%$ and opposite to the beam direction, we observed $\sim 37\%$, as shown in figure 4.3b. This means that the current geometry creates a high level of angular dependence.

Suggested geometries and Pareto optimization results. The Pareto optimization process taught us how the geometry of hollow vanadium cylinders influences the anisotropy. The results for the optimized geometries without constraints are shown in figure 4.5 and table 4.2. The main problem with the vanadium sample used in the CAMEA instrument is its large angular intensity dependence and high anisotropy. The problem addressed by Pareto optimization is how isotropic the scattering can be made, within a 60° window, while still producing sufficient intensity. The flatness is highly correlated with the sample volume and wall thickness; however, small samples are undesirable as their mean path length is small and thus produce little intensity. The multi-objective optimization search located the non-dominated solution set in a scan of 270 geometries that display sufficient flatness and intensity within the detector windows. Without beam or size constraints, the recommended geometry class is a sample 30 mm to 40 mm high with a radius of 7.5 mm to 12.5 mm and a wall thickness of 1 mm. This geometry range scatters more neutrons than the current CAMEA instrument sample while lowering the angular dependence. This range should be understood as a broad recommendation. If there are beam constraints, such as the need for a sample height and radius of 10 mm, then the recommended wall thicknesses are 0.75 mm to 1.25 mm. This geometry achieves the best combination of summed intensity and low anisotropy, according to the defined scattering metrics. The observed to expected intensity ratio could not be computed for taller samples because the flux density Ψ could not be consistently estimated. The optimization of the geometries with sample geometry constraints shown in figure 4.6 displayed a curve at the bottom with geometries that had intensity similar to the CAMEA sample, but less anisotropy. The optimized geometries are better because they show less anisotropy and the intensity is less angularly dependent. The wall thicknesses of the sample geometries on the Pareto front have thinner walls than the CAMEA sample. The observed-to-expected intensity ratio is therefore higher because the walls are thinner. This means that thicker walls might not be necessary as they do not scatter more neutrons because they re-absorb many before they can escape.

Outlook. Fixing and validating the monitor issues detailed in the thesis is an important step. The current monitor artifact issue compromises intensity detection at the poles. Correcting the asymmetry artifact and producing a spherical monitor that correctly weights and bins data is critical. Testing and comparing more complex Union geometries with McMini could further validate the methods. Using the existing McMini code base, more complex geometries could be implemented without much difficulty. It would be interesting to experimentally verify that the top-ranked geometry performs better; however, this would be an expensive procedure.

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A Theoretical background derivations

A.1 Derivation of path length sampling

Divide the sample into thin slices of area A and thicknesses dz , such that the total scattering cross section is much smaller than the area perpendicular to the beam. Essentially, it is so small that a particle cannot obscure another particle within the same slab. This is what is known as the Born approximation [1]. Now, let $p_t(z)$ be probability of transmission through the slab for a neutron. Then the probability that a neutron will be transmitted distance dz is $(1 - n\sigma dz)$. The probability of traversing dz without interaction is

$$p_t(z + dz) = p_t(z)(1 - n\sigma dz). \quad (\text{A.1})$$

Expand and move terms around to find

$$\frac{p_t(z + dz) - p_t(z)}{dz} = -p_t(z)n\sigma, \quad (\text{A.2})$$

where we recognize the LHS as the definition of a derivative.

$$\frac{dp_t(z)}{dz} = -p_t(z)n\sigma. \quad (\text{A.3})$$

This is a simple first order differential equation. Solving we find the probability of a neutron being transmitted a distance z ,

$$\int \frac{1}{p_t(z)} dp_t = -n\sigma \int_0^z dz'. \quad (\text{A.4})$$

Isolating for the probability

$$\ln(p_t(z)) = -n\sigma z, \quad (\text{A.5})$$

$$p_t(z) = \exp(-n\sigma z) = \exp(-\Sigma_t z). \quad (\text{A.6})$$

This is the expression that we are familiar with from the neutron scattering notes. The probability of an interaction in then simply

$$p_{int}(z) = 1 - \exp(-\Sigma_t z). \quad (\text{A.7})$$

We then sample a random number uniformly in $U[0, 1]$

$$p_{int}(z) = 1 - \exp(-\Sigma_t z) = U[0, 1]. \quad (\text{A.8})$$

This gives,

$$z = -\frac{\ln(1 - U)}{\Sigma_t}. \quad (\text{A.9})$$

Since U is uniformly distributed, then $1 - U$ is also uniformly distributed about a reflection around 0.5. Due to this fact we can simplify our equation to

$$z = -\frac{\ln U}{\Sigma_t}. \quad (\text{A.10})$$

Sampling from 0 to 1 will now allow us to randomly sample path lengths following the physical properties of the material.

A.2 Derivation of the expected per bin intensity

A value that is of great importance is the self-absorption. It comes from naively assuming only isotropic scattering and ignoring other effects such as multiple scattering and absorption. Start from the differential scattering cross section [1, p.13]

$$\frac{d\sigma}{d\Omega} = \frac{1 \text{ \# Neutron scattered into solid angle } d\Omega \text{ per sec.}}{\Psi}. \quad (\text{A.11})$$

This is given per atom. Now we multiply by N_{nuc} to scale up to the full sample. We also convert the number of scattered neutron to flux-like units same units as flux used in McStas: $1/(s \cdot \text{cm}^2 \cdot \text{sr} \cdot \text{energy unit})$,

$$N_{nuc} \frac{d\sigma}{d\Omega} = \frac{1}{\Psi} \frac{dI}{dt d\Omega}. \quad (\text{A.12})$$

For vanadium: $\frac{d\sigma}{d\Omega} = \frac{\sigma_{inc}}{4\pi}$,

$$N_{nuc} \frac{\sigma_{inc}}{4\pi} = \frac{1}{\Psi} \frac{dI}{dt d\Omega}. \quad (\text{A.13})$$

Rearranging the equation above,

$$N_{nuc} \frac{\sigma_{inc}}{4\pi} \Psi dt d\Omega = dI, \quad (\text{A.14})$$

where the number of nuclei is the number density times sampled volume $N_{nuc} = nV$

$$nV \frac{\sigma_{inc}}{4\pi} \Psi dt d\Omega = dI, \quad (\text{A.15})$$

where flux is given in per sec per square centimeters. So when integrating up we simply get a factor of 1 from integrating the flux. This is the equation that we use for the expected intensity in the thesis. Integrating the solid angle is slightly trickier [9]. We get

$$\Delta\Omega = \int_{\phi_1}^{\phi_2} \int_{\theta_1}^{\theta_2} \cos \theta d\theta d\phi = (\phi_2 - \phi_1)(\sin \theta_2 - \sin \theta_1), \quad (\text{A.16})$$

which for bin spanning 2 degrees by 2 degrees is $1.218 \times 10^{-3} \text{sr}$. Since flux is defined as total intensity striking the sample I_{hit} over area A we insert that and get $\frac{V}{A}$ which is simply our expected path length through sample $\langle L \rangle$. We know that number density times microscopic scattering cross section is equal to macroscopic cross section. Number density in units of cm^{-3} and microscopic cross section in units of cm^2 . Together they make the macroscopic cross section

Σ_{inc} in units of cm^{-1} Insert this into Eq. A.15 we get

$$I_{\text{bin}} = I_{\text{hit}} \Sigma_{\text{inc}} \langle L \rangle \frac{\Delta\Omega}{4\pi}, \quad (\text{A.17})$$

where we recognize that $\Sigma_{\text{inc}} \langle L \rangle$ is simply linear term for interaction probability. To get a more precise estimate we instead can use $P(\langle L \rangle) \simeq 1 - \exp(-\Sigma_{\text{inc}} \langle L \rangle)$. Thus we expect intensity in bin spanning 2 degrees by 2 degrees to be

$$I_{\text{bin}} = I_{\text{hit}} (1 - \exp(-\Sigma_{\text{inc}} \langle L \rangle)) \frac{\Delta\Omega}{4\pi}. \quad (\text{A.18})$$

The equation uses macroscopic scattering cross sections, but we can define it with more precision by accounting for all neutron interactions and then taking the fraction of neutron interactions that scatter: $\Sigma_{\text{inc}}/\Sigma_{\text{t}}$. Then we obtain

$$I_{\text{bin}} = I_{\text{hit}} \frac{\Sigma_{\text{inc}}}{\Sigma_{\text{t}}} (1 - \exp(-\Sigma_{\text{t}} \langle L \rangle)) \frac{\Delta\Omega}{4\pi}. \quad (\text{A.19})$$

Then define the probability factor as

$$P_{\text{scatt}} = \frac{\Sigma_{\text{inc}}}{\Sigma_{\text{t}}} (1 - \exp(-\Sigma_{\text{t}} \langle L \rangle)). \quad (\text{A.20})$$

We get the simple expression that tells us the expected intensity in a bin

$$I_{\text{bin}} = I_{\text{hit}} P_{\text{scatt}} \frac{\Delta\Omega}{4\pi}. \quad (\text{A.21})$$

B Supplementary validation data

B.1 Solid vanadium cylinder

Simulation Parameters. All parameters used in McStas are shown in table B.1 and McMini settings were set to match. Parameters not mentioned can be assumed to be the default values of McStas [3]

Table B.1: Parameters used in the McStas validation simulation for comparing the vanadium scattering pattern on the PSD_monitor_4PI.

Component	Parameters
Source_simple	$\lambda_0 = 4.0$, $\Delta\lambda = 0.1$, focus_xw= 0.0015, focus_yh= 0.025, dist= 1.0, radius= 0.01, $n = 10^9$
Incoherent_process	$\sigma = 5.08$, unit_cell_volume= 13.827
Union_make_material and Union_cylinder	my_absorption= 36.7, radius= 0.005, height= 0.02
PSD_monitor_4PI	nx= 360, ny= 180, radius= 1.0, restore_neutron= 1

Spatial intensity distribution comparison. To compensate for the solid angle, each bin was divided by the geometric factor $\cos \theta$ discussed in section 3.3. The corrected maps are shown in figure B.1. The poles include a higher degree of noise, since the bins are smaller and thus receive fewer rays. Once the correction is applied, the noise can blow up. 5° of the poles have been removed for this reason as well as to reduce the asymmetry problem. Then subtracting the two maps gives the difference map in figure B.1

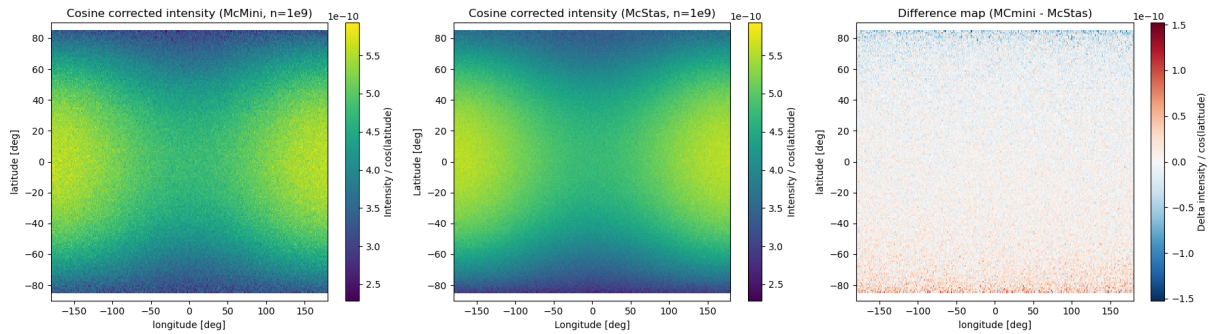


Figure B.1: Corrected intensity map and difference map of a solid cylinder with a radius of 0.5cm and height of 2cm

The difference map shows mostly noise, but there is some clear hemispheric intensity bias present. A north bias, which is believed to come from the 4π monitor is visible. This is where the asymmetry was initially discovered. Quantifying the difference map, using mean residuals, RMS residuals, mean pull, and reduced chi-square, gives the values in table B.2

Metric	Cosine-corrected map
Mean residual	$-9.328943 \times 10^{-14}$
Mean absolute residual	1.399019×10^{-11}
RMS residual	1.867373×10^{-11}
Mean pull	-1.522858×10^{-2}
Pull standard deviation	1.094898
Reduced χ^2	1.199033

Table B.2: Quantitative comparison of the cosine-corrected difference map between McMini and McStas for the solid cylinder case.

For the cosine corrected map, the mean residual (-9.33×10^{-14}) shows agreement in total intensity, while the residual RMS (1.40×10^{-11}) shows a typical mismatch per bin of the order of 10^{-11} . The pull width was 1.09 and the reduced χ^2 was 1.20.

Integrated intensities. After applying the 5° pole cut and cosine correction, the integrated intensities are

$$I_{\text{McMini}} = 2.766574 \times 10^{-5}, \quad I_{\text{McStas}} = 2.767145 \times 10^{-5}.$$

The ratio is

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} = 0.9998209,$$

Showing excellent agreement. Integrating the northern and southern hemispheres gives the intensities

	Northern Hemisphere	Southern Hemisphere	Asymmetry $A = \frac{N-S}{N+S}$
McMini	1.383150×10^{-5}	1.383425×10^{-5}	-9.944478×10^{-5}
McStas	1.404191×10^{-5}	1.362954×10^{-5}	1.490263×10^{-2}

Table B.3: Integrated intensities over northern and southern hemispheres and corresponding asymmetries for McMini and McStas simulations with a solid vanadium cylinder.

Table B.3 shows disagreement in the intensity recorded in the hemispheres of the 4π monitor and a clear asymmetry in the simulation of a solid vanadium cylinder recorded on the McStas 4π monitor. The validation performed on the solid vanadium cylinder was also performed on a solid vanadium sphere for extra validation. Interestingly, the asymmetry levels agree closely. For the solid vanadium sphere, the asymmetry levels recorded on the 4π monitor were $A_{\text{sphere}} = 1.429436 \times 10^{-2}$. For the validation of the solid vanadium sphere, see appendix B.2

B.2 Solid vanadium sphere

Raw intensity maps. The first part of the validation is a comparison between the spatial intensities of McMini and McStas. The simplest geometry to implement in C was a sphere, so the first section will be on a solid sphere. All parameters used in McStas are shown below and McMini settings were set to match. Parameters not mentioned can be assumed to be the default

Component	Parameters
Source_simple	$\lambda_0 = 4.0, \Delta\lambda = 0.1, \text{focus_xw} = 0.02, \text{focus_yh} = 0.02, \text{dist} = 1.0, \text{radius} = 0.01, n = 10^9$
Incoherent_process	$\sigma = 5.08, \text{unit_cell_volume} = 13.827$
Union_make_material and Union_sphere	$\text{my_absorption} = 36.7, \text{radius} = 0.01$
PSD_monitor_4PI	$\text{nx} = 360, \text{ny} = 180, \text{radius} = 1.0, \text{restore_neutron} = 1$

Table B.4: Simulation parameters for the spherical vanadium sample.

values of McStas [3] The raw monitor data from McMini and McStas is seen in Fig B.2. At first look they are indistinguishable. Both exhibit the expected angular dependence and a decrease of intensity towards the poles.

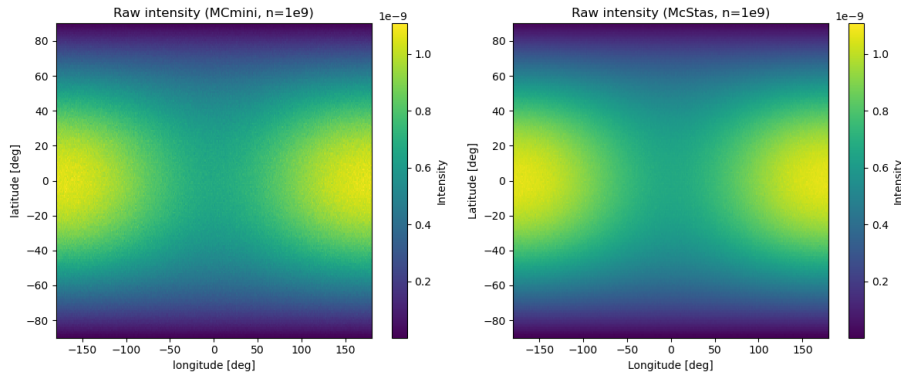


Figure B.2: Raw intensity map of a solid sphere with radius of 1cm

The total intensities obtained from both simulations are

$$I_{\text{McMini}} = 3.6189 \times 10^{-5}, \quad I_{\text{McStas}} = 3.6195 \times 10^{-5}.$$

The ratio is

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} = 0.99984,$$

demonstrating excellent normalization agreement at the 10^{-4} level.

Difference map To correct for solid angle, each bin was divided by the geometrical $\cos(\text{latitude})$ factor. The corrected maps are shown in Fig. B.3. The poles include a higher degree of noise, since the bins are smaller and thus receive fewer rays. Once the correction is applied this noise can blow up. Therefore the data within 5 degrees of the poles has been removed. Then subtracting the two maps gives the difference map in figure B.3

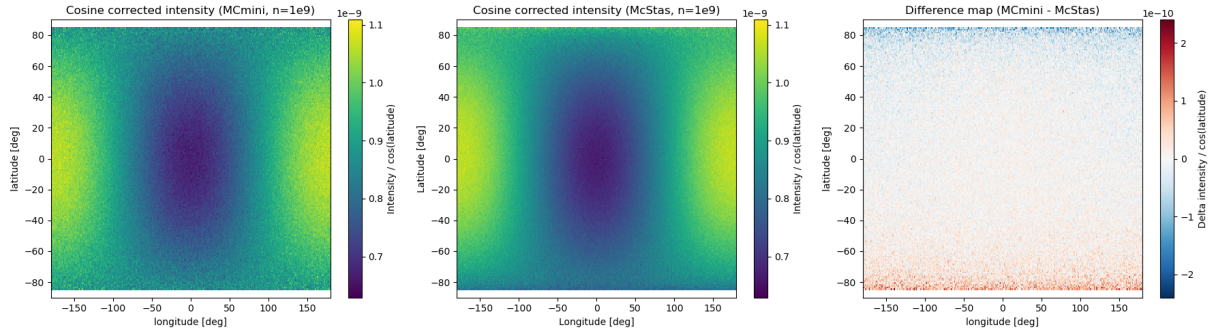


Figure B.3: Corrected intensity map and difference map of a solid sphere with a radius of 1cm

The difference map shows mostly noise, but some clear bias is visible. A north bias, which is believed to come from the 4π monitor, shows itself. This is where the asymmetry was initially discovered. Quantifying to difference map, using mean residuals, RMS residuals, mean pull, and reduced chi-square gives the values in table B.5

Metric	Cosine-corrected map
Mean residual	$-1.562070 \times 10^{-13}$
Mean absolute residual	2.216481×10^{-11}
RMS residual	3.149105×10^{-11}
Mean pull	-5.172632×10^{-3}
Pull standard deviation	1.142612
Reduced χ^2	1.305589

Table B.5: Quantitative comparison of the cosine-corrected difference map between McMini and McStas.

For the cosine corrected map the mean residual (-1.56×10^{-13}) shows agreement in total intensity, while the residual RMS (3.15×10^{-11}) shows a typical per bin mismatch on the order of 10^{-11} . The pull width was 1.14 and the reduced χ^2 was 1.31, showing some excess over pure statistical noise. This excess being the asymmetry that is visible, and not enough to say that the simulations do not match.

Equator and meridian cut Taking a cut along the equator and meridian shows the agreement along a line of bins. The result is seen in figure B.4

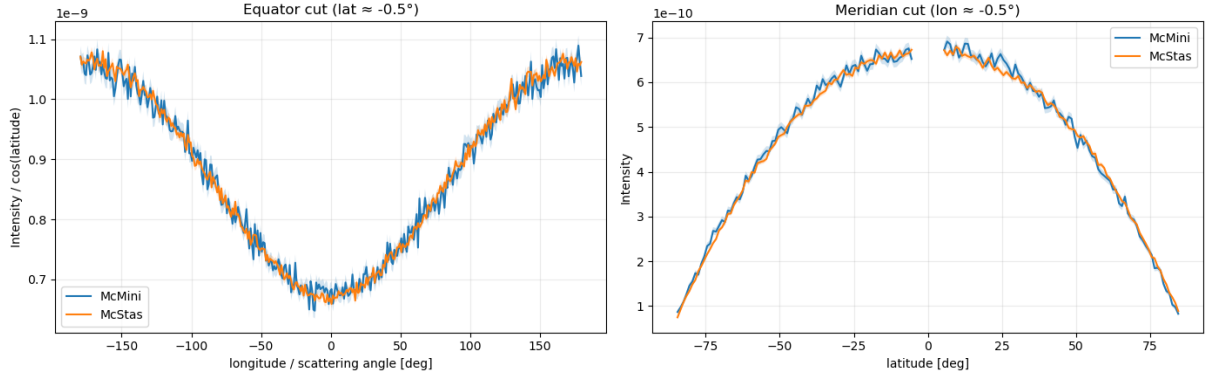


Figure B.4: Equator and meridian cut for a solid sphere with a radius of 1cm

The cuts show excellent agreement for both the equator and meridian cut. McMini displays greater levels of noise, but this is because it does not use the same variance reduction method that McStas uses.

Integrated intensities After applying the 5° pole cut and the cosine correction, the total integrated intensities of the full 4π maps are

$$I_{\text{McMini}} = 5.372803 \times 10^{-5}, \quad I_{\text{McStas}} = 5.373468 \times 10^{-5}.$$

The ratio is

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} = 0.9998762,$$

Integrating the northern and southern hemisphere.

$$N_{\text{McMini}} = 2.686408 \times 10^{-5}, \quad S_{\text{McMini}} = 2.686395 \times 10^{-5},$$

and

$$N_{\text{McStas}} = 2.725139 \times 10^{-5}, \quad S_{\text{McStas}} = 2.648329 \times 10^{-5}.$$

This corresponds to hemisphere asymmetries

$$A_{\text{McMini}} = \frac{N - S}{N + S} = 2.484464 \times 10^{-6}, \quad A_{\text{McStas}} = \frac{N - S}{N + S} = 1.429436 \times 10^{-2}.$$

showing a clear asymmetry for the hemisphere, while the total intensity agrees excellently.

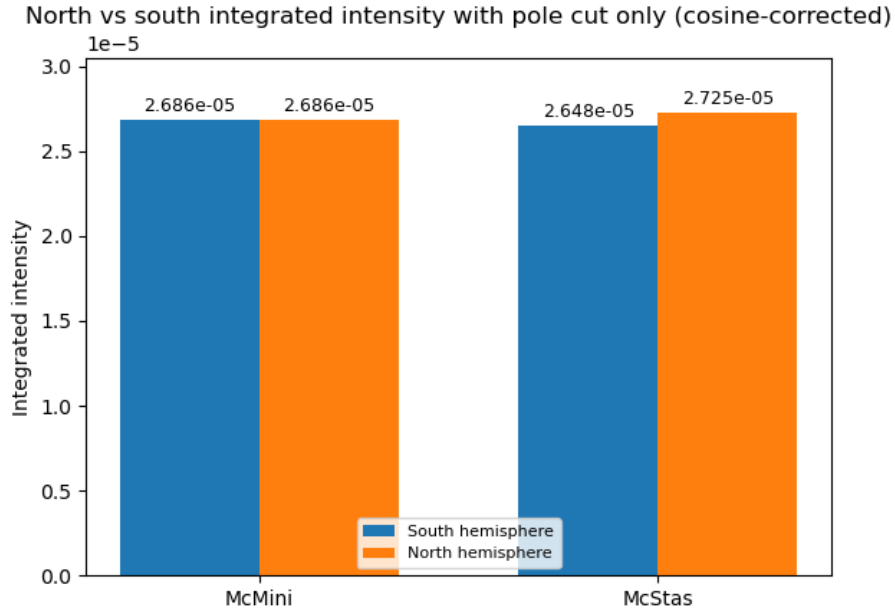


Figure B.5: Cosine-corrected integrated intensities in the northern and southern hemispheres

A clear asymmetry is present in the McStas result, whereas the McMini result remains essentially symmetric. As was discussed in section C.2, asymmetry is present in the 4π monitor. The largest part of the asymmetry observed here, seems to come from that issue.

Scatter-order analysis Last part of the validation for the solid sphere will be on intensity maps of neutrons with different scatter orders. McMini has been made with the ability to count interactions just like McStas. figure B.6 shows the intensity maps of neutrons with scatter order equal to 1, 2, 3, 4, and ,5.

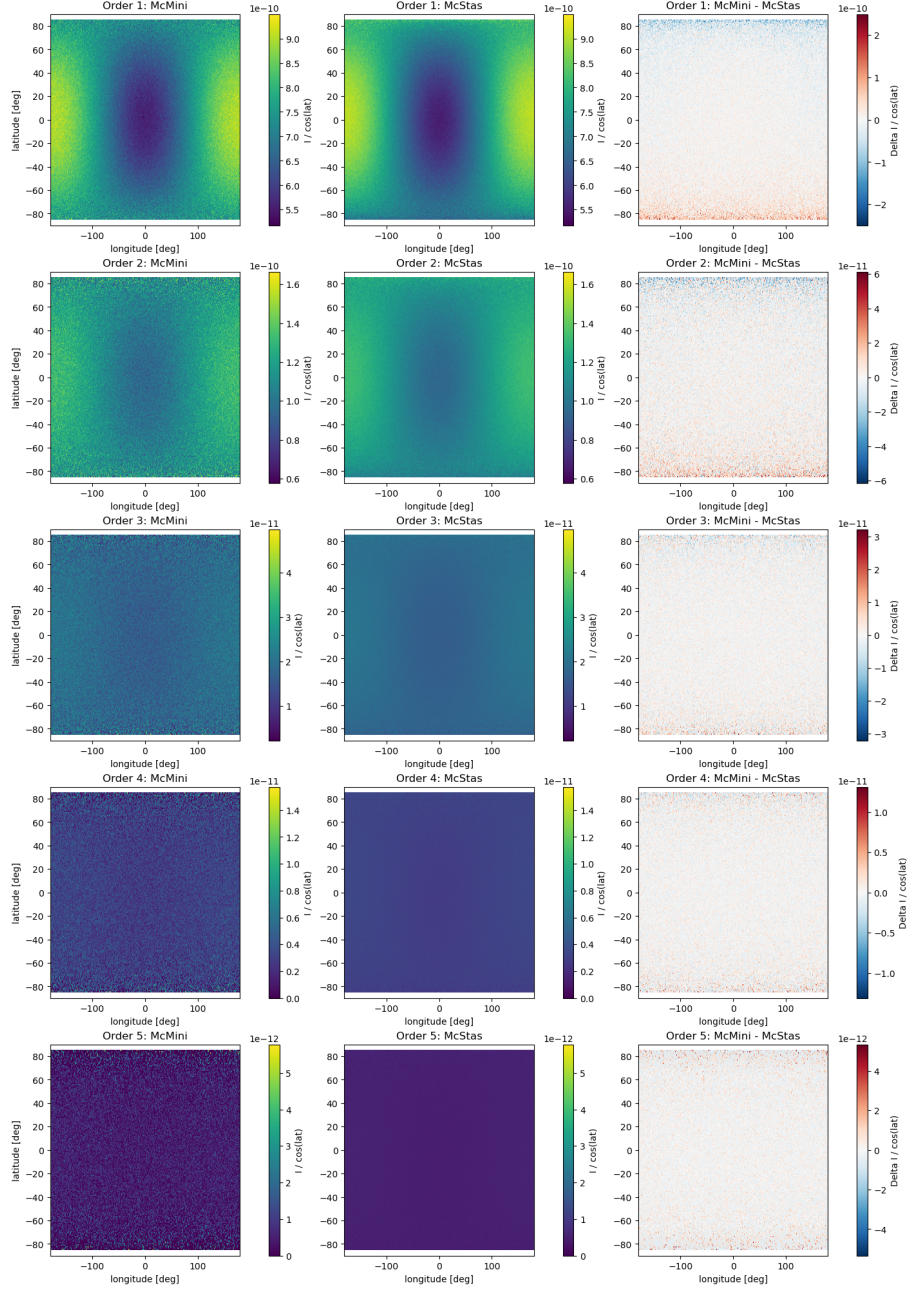


Figure B.6: Scatter order maps for McStas and McMini, together with their difference map

Visually, all scatter orders match great. For high orders, such as 5 the higher noise in McMini due to no variance reduction becomes very clear. Table B.6 shows integrated intensity and intensity ratios of different scatter orders. It also shows the percentage share that a scatter order makes up of the total intensity. For all orders, integrated intensity are in unity. They agree on the share of intensity, and RMS shows low per-bin disagreement. The asymmetry is still present, but seems to be less for higher orders.

Table B.6: Cosine-corrected, order-resolved integrated intensities for the sphere case with pole cut $|\text{latitude}| \leq 85^\circ$. Shares are given relative to the summed intensity over orders 1–5.

Order	I_{McMini}	$\text{Share}_{\text{McMini}}$	I_{McStas}	$\text{Share}_{\text{McStas}}$	$I_{\text{McMini}}/I_{\text{McStas}}$	RMS
1	4.536851×10^{-5}	84.45%	4.537335×10^{-5}	84.45%	0.99989	2.82×10^{-11}
2	6.990619×10^{-6}	13.01%	6.991936×10^{-6}	13.01%	0.99981	8.61×10^{-12}
3	1.138327×10^{-6}	2.12%	1.139436×10^{-6}	2.12%	0.99903	3.22×10^{-12}
4	1.912518×10^{-7}	0.36%	1.908454×10^{-7}	0.36%	1.00213	1.30×10^{-12}
5	3.250116×10^{-8}	0.06%	3.241492×10^{-8}	0.06%	1.00266	5.30×10^{-13}

C Technical note on the 4π monitor asymmetry

C.1 Structural analysis on the cause of the observed asymmetry

This asymmetry artifact was initially discovered during the multiple scattering simulation with spherical and cylindrical samples. This caused some confusion since it was unexpected that symmetrical sample would yield asymmetrical results. This problem was investigated and has been narrowed down to a PSD_monitor_4PI issue. A brief mention of other factors that were investigated since they could have caused the issue included:

1. **Incorrect monitor selection / inclusion of unscattered rays, and beam-stop treatment.** The possibility that the asymmetry came from the transmitted neutron, or from incorrect beam-stop handling was tested by restricting the analysis to only scattered neutrons, i.e. using $n_{\text{reflections}} > 0$ in McStas and McMini. The asymmetry persisted so was not caused by this.
2. **Cosine correction of the 4π monitor.** The monitor maps were compared before and after correction for the solid angle. The asymmetry was already present in the raw data, so the effect is not caused by the cosine correction.
3. **Monitor geometry in McMini.** The McMini spherical monitor was tested in two modes. geometric sphere-hit binning and direction-only binning. This changed some details, but the discrepancy could not be explained as a geometric artifact on the McMini side.
4. **Incorrect uncertainty model.** The McMini monitor was extended to record weighted uncertainties using $\sqrt{\sum w_i^2}$, instead of a Poisson assumption. This improved statistics but did not reduce asymmetry.
5. **Source asymmetry on a flat monitor** As was already shown, no asymmetry showed itself for a flat monitor. This includes a flat monitor immediately after the source and downstream. With a north to south asymmetry at the level of 10^{-6} to 10^{-5} and a negligible fitted vertical slope
6. **Absorption / wavelength-dependent attenuation.** The scattering experiment was repeated in McStas with absorption set to 0. The asymmetry remained.
7. **Scatter-order dependence.** The asymmetry remained for one, two, and three scattering events. The asymmetry was strongest for the single scattering event and was reduced to mostly noise for higher order.

8. **Simple indexing or sign-convention errors.** Tests involving latitude flips, longitude flips, and ± 1 -bin shifts did not remove the residual structure.
9. **Bare 4π monitor response without a sample.** This is what was shown in figure C.1, and seems to be the main cause of asymmetry. This configuration agrees in total intensity but favors a north to south bias.

Thus, the asymmetry is not caused by inclusion of unscattered rays, beam stop handling, co-sine correction, source non uniformity, absorption, or a simple indexing mistake. The evidence points to an issue in the McStas 4π monitor, which favors a north bias.

C.2 Asymmetry analysis on the flat monitor vs. spherical monitor

Comparison with McStas flat monitor. To verify the flat monitor implemented in McMini, a comparison was made with an instrument in McStas that contained only a source and a flat PSD_monitor was made. Instrument parameters used in McStas and matched in McMini were:

Component	Parameters
Source_simple	$\lambda_0 = 4.0$, $\Delta\lambda = 0.1$, focus_xw=focus_yh= 0.5, dist= 1.0, radius= 0.25, $n = 10^9$
PSD_monitor	xwidth=yheight= 1.0, nx=ny= 2000, restore_neutron= 1

Table C.1: Simulation parameters for the source and planar PSD monitor.

The comparison will include visual inspection, their difference map, and a check for asymmetry. Since the sources are equal we expect only noise to be left when subtracting the two, and since it is symmetric, we would expect only noise when mirroring the south onto the north side. Raw data and the difference map can be seen in figure C.1.

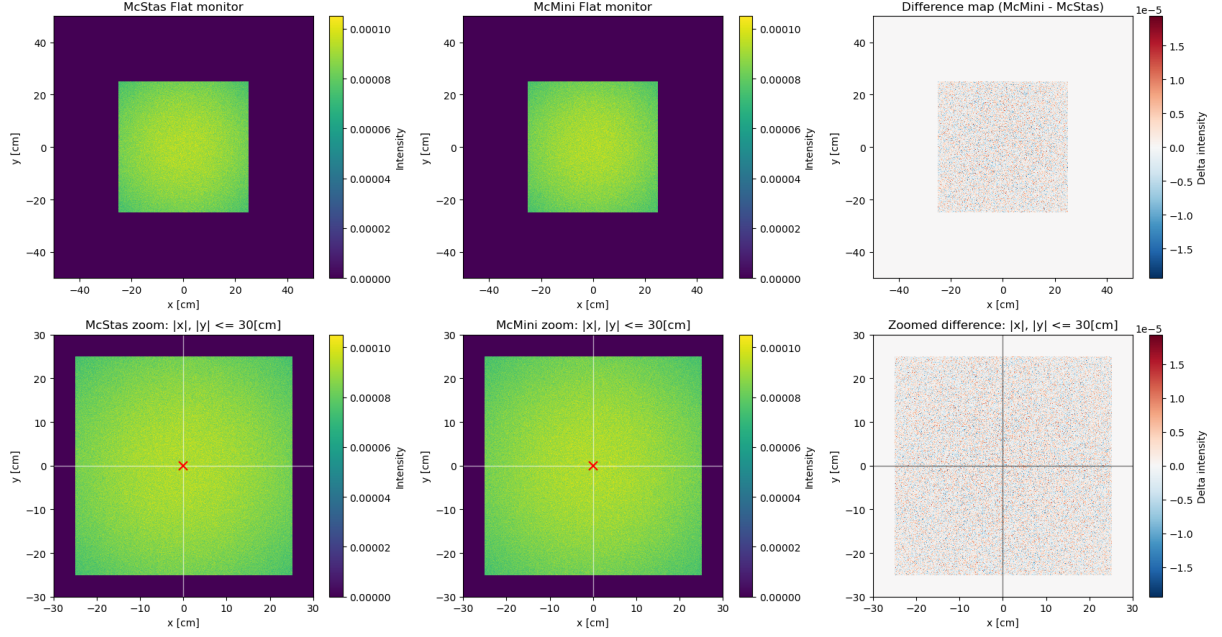


Figure C.1: Comparison and zoomed comparison of McStas and McMini

The same color scale has been forced as is standard practice. Visually they are indistinguishable, and then subtracting the two leaves nothing but noise on the difference map.

Flat monitor symmetry diagnostics. Since the geometry is symmetric, no north to south or east to west asymmetry is expected. To verify that no artificial directional bias is introduced at the source or detection level, symmetry tests were performed by mirroring the data onto itself.

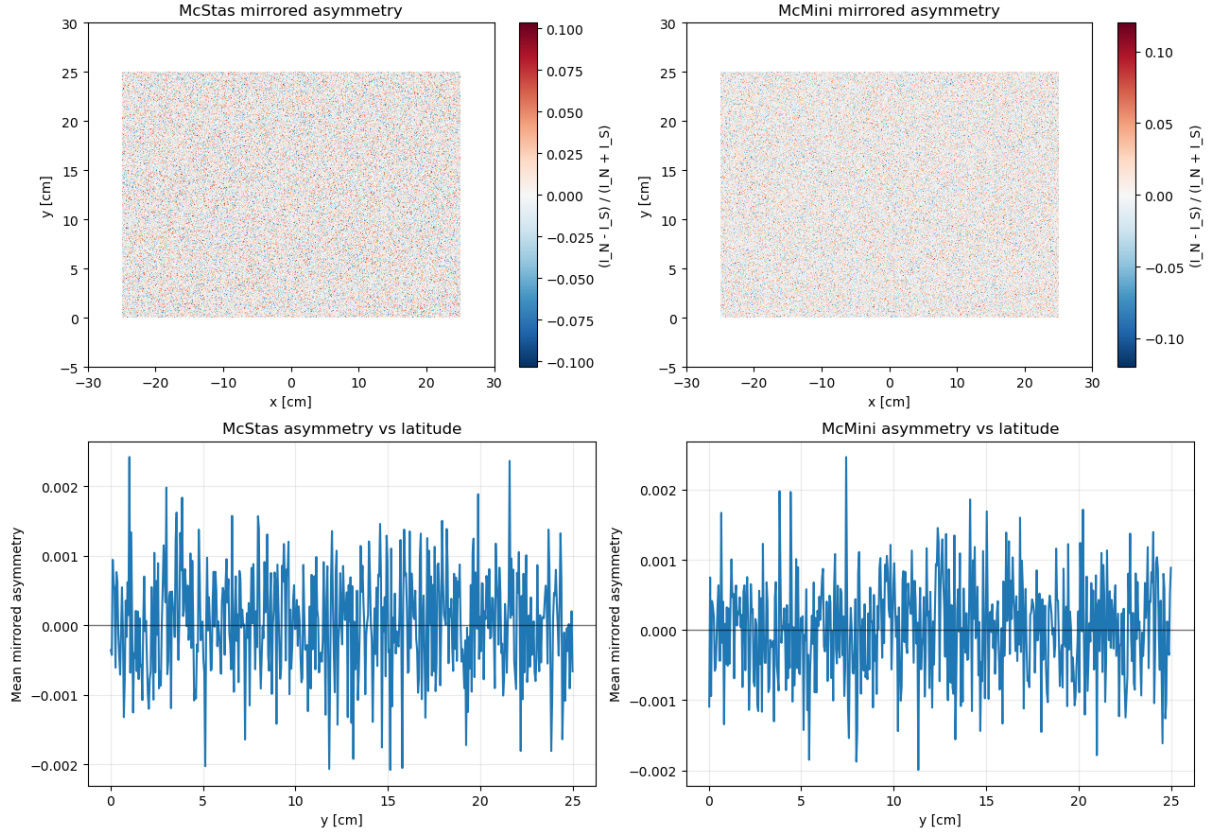


Figure C.2: Asymmetry analysis of McStas and McMini

Mirroring the data onto itself shows no signs of asymmetry, and both mean data along each axis show only noise around 0.

Integrated intensity. Quantifying further and integrating the total intensity together with integrating the north-south and east-west side,

$$I_{\text{tot}} = I_N + I_S, \quad A = \frac{I_N - I_S}{I_N + I_S}.$$

Quantity	McStas	McMini
Total intensity I_{tot}	8.604235×10^1	8.604149×10^1
North-south asymmetry $(N - S)/(N + S)$	3.08×10^{-5}	9.81×10^{-6}
East-west asymmetry	-9.87×10^{-6}	2.48×10^{-5}

Table C.2: Integrated intensity comparison for the flat monitor.

Both implementations agree excellently on the total intensity down to the 10^{-2} , and both implementations exhibit symmetry to within $\mathcal{O}(10^{-5})$, with no systematic north-south or east-west bias. This confirms that neither the source model nor the flat monitor introduces any significant bias. Any asymmetry in later 4π monitor results must therefore originate from angular binning or scattering/recording chain rather than from the source itself.

Comparison with McStas 4π monitor. To verify the 4π monitor, again a comparison with an instrument containing only a source and a monitor was made. This time the monitor is a PSD_monitor_4PI with the parameters in table B.5: The comparison includes the same metrics

Component	Parameters
Source_simple	$\lambda_0 = 4.0, \Delta\lambda = 0.1, \text{focus_xw}=\text{focus_yh}= 0.5,$ $\text{dist}= 1.0, \text{radius}= 0.25, n = 10^9$
PSD_monitor_4PI	$\text{radius}= 1.0, \text{nx}= 3600, \text{ny}= 1800,$ $\text{restore_neutron}= 1$

Table C.3: Simulation parameters for the source and 4PI PSD monitor.

used in the flat monitor inspection. Raw data and the difference map can be seen in figure C.3.

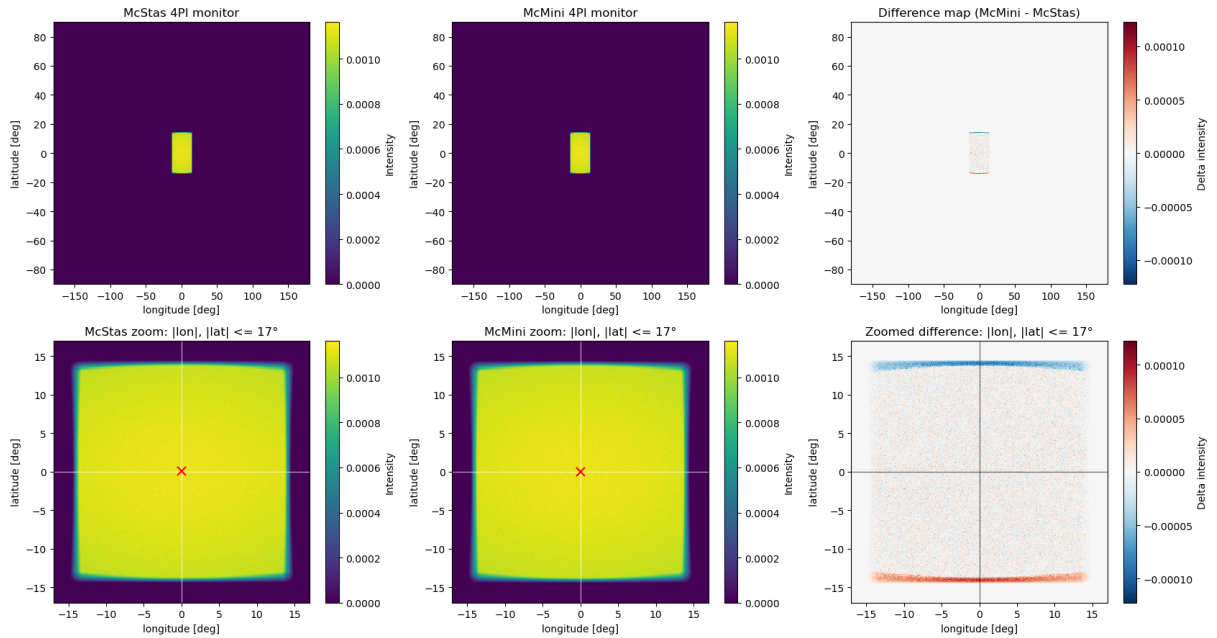


Figure C.3: Comparison, zoomed comparison and difference map of direct beam on a 4π monitor.

The same color scale had again been forced. Visually they are indistinguishable, however, this time when subtracting the data generated in McStas from that generated in McMini there is something left other than just noise. There is a clear big negative difference and a clear positive difference located at the top and bottom of the data.

4π monitor symmetry diagnostics. This asymmetry artifact was initially discovered during the multiple scattering simulation with spherical and cylindrical samples. This caused some confusion since it was unexpected that symmetrical sample would yield asymmetrical results. This problem was investigated and has been narrowed down to a PSD_monitor_4PI issue. Flipping the south hemisphere onto the north hemisphere we see where the greatest discrepancy is. This was done for both McStas and McMini in figure C.4

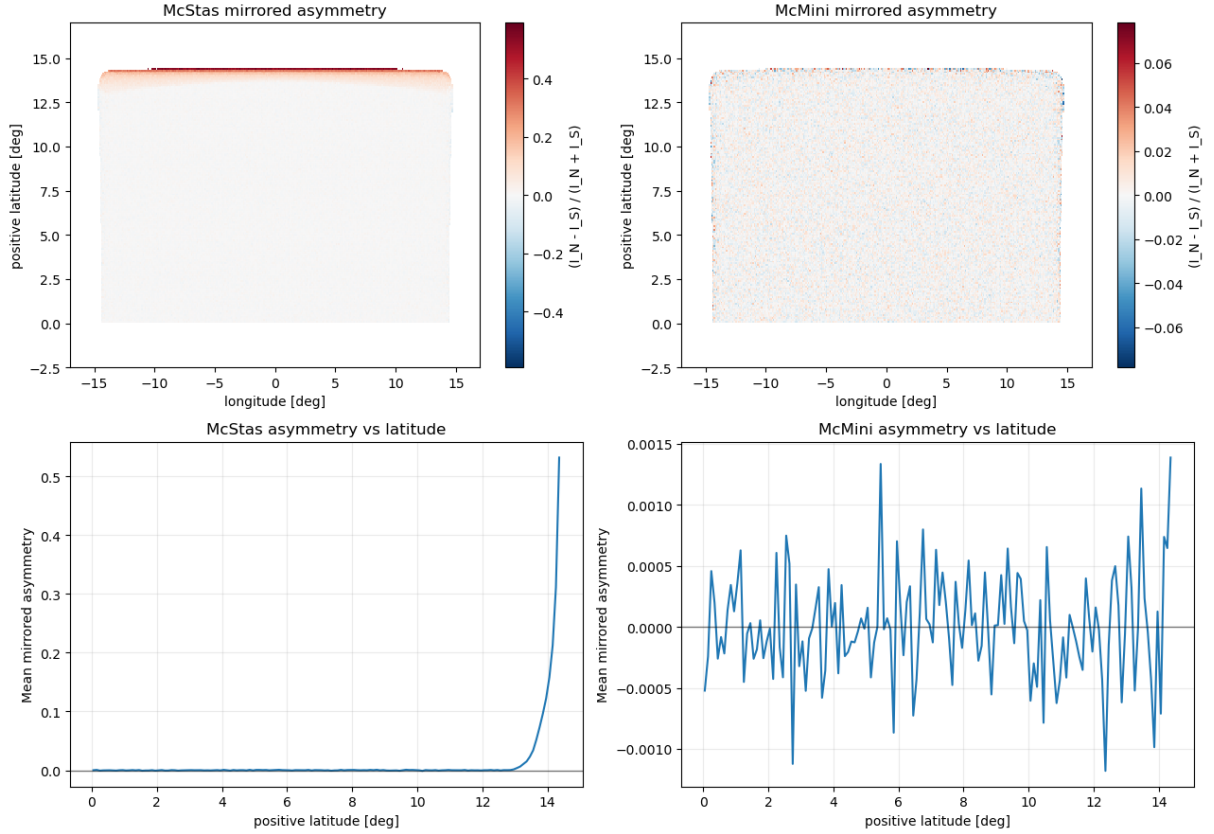


Figure C.4: Mirrored map and asymmetry vs latitude

The figure shows a major asymmetry located at the edge in the McStas simulation. Asymmetry vs latitude shows exponential growth towards the edge at 14° . For McMini the asymmetry is essentially zero, and only noise is left for the mirrored map and asymmetry vs latitude.

Integrated intensity. Table C.4 shows the integrated intensity values. It is again seen that the total intensities agree to a very high degree, but McStas shows a north to south asymmetry of 0.37%, whereas McMini agree down to the 10^{-6} .

Quantity	McStas	McMini
Total intensity I_{tot}	8.604208×10^1	8.604149×10^1
North hemisphere N	4.317913×10^1	4.302105×10^1
South hemisphere S	4.286295×10^1	4.302044×10^1
North-south asymmetry $(N - S)/(N + S)$	3.674779×10^{-3}	7.121273×10^{-6}

Table C.4: Integrated intensity comparison for the 4π monitor.

Diagnostics with a 4π source A test with a 4π source spreading equally in all directions showed similar results of north-south asymmetry. This diagnostics is cleaner because we would expect a completely uniform spread on the monitor.

The comparison and the result of subtracting the southern hemisphere from the northern hemisphere is shown in figure C.5. Computing the p-value for the McStas north-south asymmetry

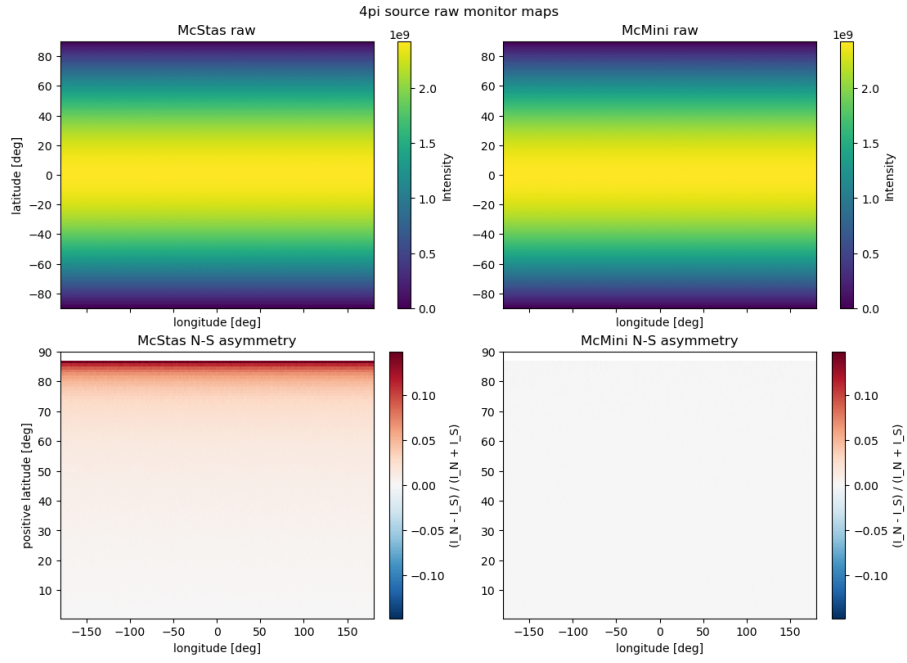


Figure C.5: A comparison of the data recorded from a 4π source onto the spherical monitor and the southern hemisphere mirrored onto the northern hemisphere.

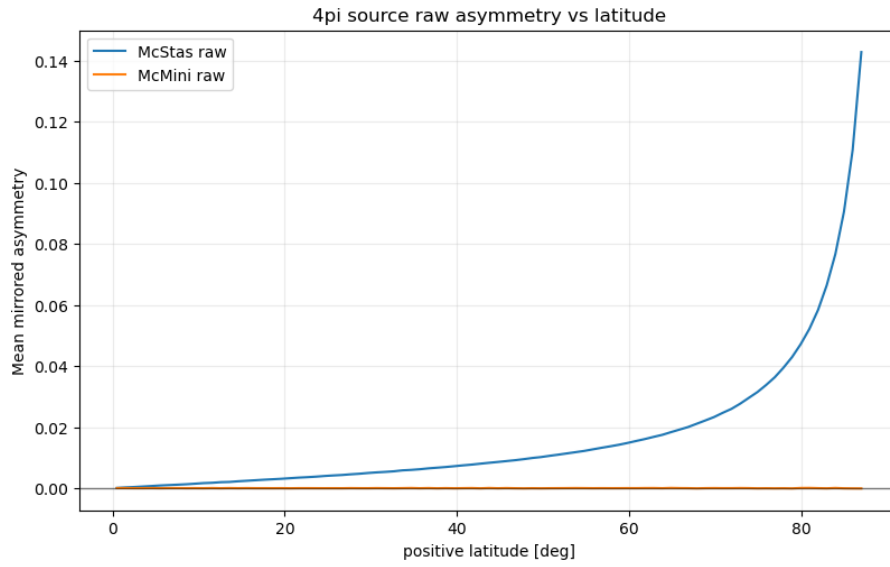


Figure C.6: The degree of asymmetry as a function of latitude

gives 0, while McMini give 0.32975. Figure C.6 shows how much the asymmetry increases as a function of latitude.

D CAMEA instrument at PSI

D.1 CAMEA instrument default parameters

Parameter	Value	Description
SourceE	5 meV	Mean source neutron energy
DeltaSourceE	0.5 meV	Width of source energy spectrum
EI	5 meV	Incoming neutron energy selected by the monochromator
A3	0°	Sample rotation angle
A4	−45°	Sample take-off angle / tank angle
SAMPLE	0	Sample type, where 0 corresponds to vanadium
RV_mono_Bool	1	Vertical monochromator focusing enabled
RH_mono_Bool	1	Horizontal monochromator focusing enabled
Lv	1.872 m	Vertical focusing distance
Lh	1.340 m	Horizontal focusing distance
monoShieldingWidth	0.062 m	Width of monochromator shielding opening
monoShieldingHeight	0.14 m	Height of monochromator shielding opening
monoShieldingDistance	0.235 m	Distance between monochromator and shielding opening
MonoShielding	1	Monochromator shielding enabled
Be_filter	1	Beryllium filter enabled
sampleHeight	0.01 m	Sample height
sampleRadius	0.01 m	Sample radius
sampleThickness	0.0026 m	Wall thickness of hollow cylindrical sample
samplePosY	0.00 m	Vertical offset of sample
sampleFluxDiagnosticXWidth	0.08 m	Width of diagnostic flux monitor
sampleFluxDiagnosticYHeight	0.08 m	Height of diagnostic flux monitor

Table D.1: Relevant simulation parameters for the PSI CAMEA instrument setup.

D.2 Parameter values for the geometry scan

The geometry scan discussed in section 4.4 was created using the following shell script: All sample dimensions are given in millimeters in the loops below but are converted to meters before being supplied to McStas. This scan goes over 9 heights, 5 radii and 6 wall thicknesses for a total of 270 simulations. The simulations were each run with 2×10^8 neutron rays and 8 MPI processes.

Table D.2: Parameter values used in the geometry scan.

Parameter	Values
Sample height h mm	5, 10, 15, 20, 25, 30, 35, 40, 50
Sample radius r mm	5, 7.5, 10, 12.5, 15
Wall thickness t mm	0.5, 1.0, 1.5, 2.0, 2.5, 3.0
Incident energy E_i [meV]	5.0
Energy spread ΔE [meV]	0.5
Number of neutron rays	2×10^8
MPI processes	8
Number of simulations	270

Listing D.1: Shell script used for the McStas geometry scan.

```

for h_mm in 5 10 15 20 25 30 35 40 50; do
  for r_mm in 5 7.5 10 12.5 15; do
    for t_mm in 0.5 1.0 1.5 2.0 2.5 3.0; do

      h=$(python -c "print($h_mm/1000)")
      r=$(python -c "print($r_mm/1000)")
      t=$(python -c "print($t_mm/1000)")

      dir="scan_h${h_mm}mm_r${r_mm}mm_t${t_mm}mm"

      mcrun FullInstrument_v5_working.instr \
        -n 2e8 --mpi=8 \
        --dir "$dir" \
        SourceE=5.0 DeltaSourceE=0.5 EI=5.0 \
        A3=0 A4=-45 SAMPLE=0 \
        RV_mono_Bool=1 RH_mono_Bool=1 \
        Lv=1.872 Lh=1.340 \
        monoShieldingWidth=0.062 \
        monoShieldingHeight=0.14 \
        monoShieldingDistance=0.235 \
        MonoShielding=1 \
        Be_filter=1 \
        sampleHeight="$h" \
        sampleRadius="$r" \
        sampleThickness="$t" \
        samplePosY=0.00 \
        sampleFluxDiagnosticXWidth=0.08 \
        sampleFluxDiagnosticYHeight=0.08

    done
  done
done

```

KU's AI-deklaration

Deklaration for anvendelse af generative AI-værktøjer

☒ Jeg/vi har benyttet generativ AI som hjælpemiddel/værktøj (sæt kryds)

☐ Jeg/vi har **IKKE** benyttet generativ AI som hjælpemiddel/værktøj (sæt kryds)

Hvis brug af generativ AI er tilladt til eksamen, men du ikke har benyttet det i din opgave, skal du blot krydse af, at du ikke har brugt GAI, og behøver ikke at udfylde resten.

Oplist, hvilke GAI-værktøjer der er benyttet, inkl. link til platformen (hvis muligt):

ChatGPT: <https://chatgpt.com/>

Codex: <https://marketplace.visualstudio.com/items?itemName=openai.chatgpt>

Beskriv hvordan generativ AI er anvendt i opgaven:

1) Formål:

Formål med brug af GAI har været som en sparringpartner igennem projektet. Jeg har brugt det til kilde-/formaterings-hjælp i Overleaf Latex, samt overordnet passende struktur af latexdokumentet til projektet. GAI er blevet brugt som et hjælpeværktøj til at debugge mit program, McMini, skrevet i C.

2) Arbejdsfase:

GAI er blevet anvendt løbende igennem arbejdsfasen af projektet. Det er blevet anvendt i sidste uge af projektet hvor jeg har læst igennem for stavefejl, grammatik og tegnsætning.

3) Anvendelse af output:

Intet direkte output fra GAI er blevet brugt. GAI er blevet brugt til at finde stavefejl og grammatik fejl. Intet fagligt indhold er blevet ændret som resultat af disse rettelser.